

HalluPrism: When Multimodal Uncertainty Should Diagnose, Not Decide

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Abstract

Multimodal Large Language Models (MLLMs) can give similar confidence to answers that fail for different reasons: the image may be unclear, the question may invite a plausible answer irrespective of the image, or the model may bind the wrong image-text relation. To address this, we propose HalluPrism, a behavioral diagnostic for MLLM uncertainty. Given an image, prompt, and answer, HalluPrism re-runs the answer after visual degradation, blank-image replacement, and grounding or relation checks. These probes record answer changes under targeted input stress, producing a signature over visual fragility (V), language-prior reliance (L), and cross-modal alignment instability (A). Across 58K+ examples from HallusionBench, POPE, VSR, and VizWiz-VQA with four MLLMs, we find four patterns. First, the raw signature does not equal diagnostic value. We find language-prior reliance is most common, while alignment instability better separates failures. Second, the signals overlap and should be read jointly. Third, controlled analyses show that the signature improves failure-type prediction: XGBoost AUROC rises from 0.78 with scalar confidence to 0.95 with (V, L, A) and 0.97 with confidence added. Fourth, forcing the signature into one abstention score harms the correctness ranking. These results suggest that multimodal uncertainty should be used to diagnose failure structure before abstention or correction.

1 Introduction

A visually-grounded answer can be unsafe even when a Multimodal Large Language Model (MLLM) assigns high confidence to it. Consider a street-scene image where a traffic cone is to the right of a parked car, and the user asks, “*Is the traffic cone to the left of the car?*” The model may answer “yes” with high confidence. This single confident answer can hide different risks. The cone

may be small, blurred, or partly occluded, suggesting weak visual evidence. The model may still answer “yes” after the image is replaced with a blank canvas, pointing to language-prior reliance on the question alone. It may also recognize both the cone and the car while reversing the left-right relation, suggesting cross-modal binding failure. In such cases, while scalar uncertainty can help decide whether the model should abstain, it cannot explain the failure source.

Existing hallucination benchmarks already show that multimodal errors are heterogeneous, covering object hallucination, visual illusion, attribute error, spatial error, and answerability failure (Li et al., 2023; Guan et al., 2024; Wang et al., 2023; Liu et al., 2023c). Confidence, calibration, and selective prediction methods usually rank likely correctness or decide whether the model should abstain (Chow, 1970; Kadavath et al., 2022; Kuhn et al., 2023; Geng et al., 2023), while recent MLLM uncertainty work moves closer to hallucination detection through evidential conflict, hidden-state reliability, and uncertain visual tokens (Huang et al., 2025; Mushtaq et al., 2025; Seo et al., 2025). However, scalar abstention is not an internal model guarantee. It is rather a deployment guardrail whose validity depends on designer-chosen scores, thresholds, validation sets, and risk assumptions that may fail for new open-world instances. We, therefore, argue that diagnosis should be the core object of multimodal uncertainty — before answering or correcting, the system should identify *whether risk comes from weak visual evidence, language-prior reliance, or cross-modal binding*.

We propose HalluPrism, a source-signature diagnostic for MLLM uncertainty. Given an image, a prompt, and a generated answer, HalluPrism computes scalar confidence c and a source signature $s(x) = (V, L, A)$ for visual fragility (V), language-prior reliance (L), and cross-modal alignment instability (A). In the traffic-cone exam-

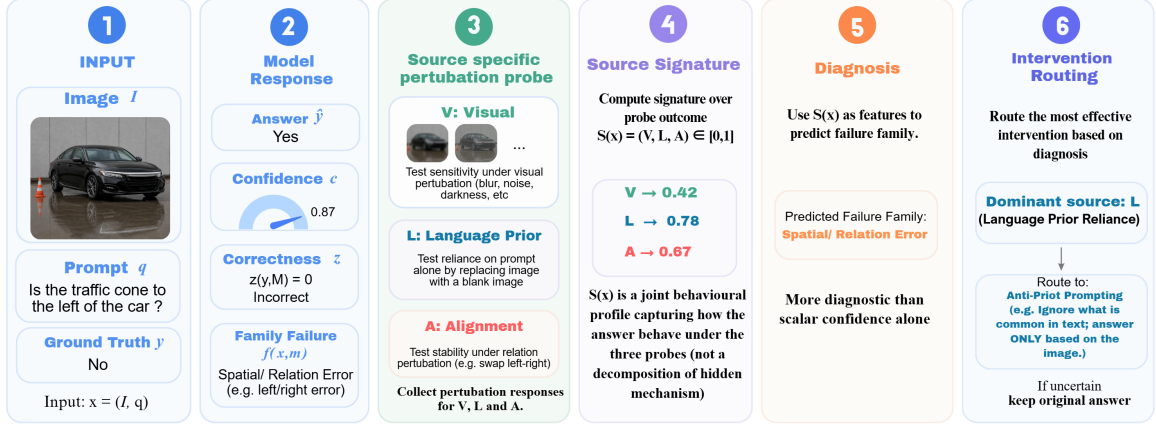


Figure 1: **An overview of the HalluPrism perturbation-response source-signature protocol.** The scores are behavioral probes, not a causal decomposition. Full prompt templates and probe parameters are given in Appendix C.

ple, V increases when blur, crop, darkness, or noise changes the answer or its confidence, L increases when the answer survives blank-image replacement, and A increases when grounding or relation perturbations expose instability in the left-right binding. The signature $s(x)$ is a perturbation-response profile, not a decomposition of hidden mechanisms. Hence, a high L does not prove that the model never used the image, while high V or A records sensitivity rather than an isolated internal cause. We use HalluPrism as a *joint* behavioural signature for diagnosis and correction-path selection. It first acts as a diagnostic feature for failure-family prediction. It then selects a correction path for risky predictions: visual fragility triggers *visual re-grounding* of the cone and car, language-prior dominance triggers *anti-prior prompting* that asks the model to ignore question-only plausibility, and alignment sensitivity triggers *relation checking* of whether the cone is to the left or right of the car. We call this source-conditioned choice of correction prompt *intervention routing*.

Across 58,440 examples from Gemma 4, LLaVA-Mistral/Vicuna, and Qwen3-VL on HallucinationBench, POPE, VSR, and VizWiz-VQA, we find four patterns. First, raw prevalence does not have diagnostic value. Although L dominates 99.7% of samples, A is seen to be the strongest tree-based feature for failure-family prediction. Second, the scores are not independent labels. If V , L , and A were indeed source-specific, then visual, language, and alignment-ruined inputs would have mainly moved corresponding matching coordinates. However, across $4 \times 4 \times 3$ ruined-condition checks, only 18/48 such events pass. Visual inputs pass

in 11/16 settings, while language and alignment inputs pass in 2/16 and 5/16. Thus, (V, L, A) is a joint source signature, not independent causes. Third, this joint signature remains highly diagnostic. In a controlled setup where the failure-type classifier is given the regime information (i.e., dataset-model) of the answer, we observe XGBoost AU-ROC improving from 0.78 with scalar confidence to 0.95 with (V, L, A) , and to 0.97 with confidence added. Fourth, the signature guides safer correction. Matched routing selects visual re-grounding, anti-prior prompting, or relation checking, instead of one generic prompt for all examples. On Qwen3-VL/VSR, generic prompting drops accuracy from 87.4% to 76.2% and breaks 15.5% of correct answers. Matched routing maintains 87.6% accuracy and lowers breakage to 2.7%.

Contributions. (i) We argue that source-signature-based diagnosis should be the core of multimodal uncertainty. (ii) We propose HalluPrism, a perturbation-response signature over visual fragility, language-prior reliance, and cross-modal alignment instability. (iii) We show that controlled probes expose interventional entanglement rather than clean source separability. (iv) We show that this entangled signature improves failure-family diagnosis, while naive scalarization can hurt abstention ranking.¹

2 Related Work

Hallucination Evaluation. Vision-language hallucination work shows that multimodal errors are heterogeneous. Captioning and Visual Question

¹Our source code is attached as supplementary and will be open-sourced upon acceptance of the paper.

Answering connect object hallucination and visual mis-reasoning to language priors and weak grounding (Rohrbach et al., 2018; Goyal et al., 2017). Benchmarks such as POPE, HallusionBench, AMBER, MME, and MMBench make this explicit across object presence, visual illusion, attributes, relations, perception, reasoning, and instruction following (Li et al., 2023; Guan et al., 2024; Wang et al., 2023; Fu et al., 2023; Liu et al., 2023c). HalluPrism instead measures a behavioral source signature over visual fragility, language-prior reliance, and cross-modal alignment instability.

Uncertainty and Abstention. Reject-option and selective-classification work treats uncertainty as a decision object. A model answers when risk is low and abstains otherwise (Chow, 1970; Geifman and El-Yaniv, 2017, 2019). Calibration asks whether confidence matches empirical correctness (Guo et al., 2017). Language-model uncertainty uses self-estimation, token probabilities, semantic equivalence, and sample inconsistency as reliability signals (Kadavath et al., 2022; Kuhn et al., 2023; Geng et al., 2023; Manakul et al., 2023). Multimodal uncertainty adds visual abstention, evidential conflict, hidden-state reliability, and visual-token uncertainty (Eisenschlos et al., 2024; Srinivasan et al., 2024; Huang et al., 2025; Mushtaq et al., 2025; Seo et al., 2025). Our critique is narrower. Abstention has statistical value under its assumptions. However, a thresholded score is warranted by validation transfer and need not diagnose risk in a new open-world instance.

Probing and Attribution. Behavioral testing studies model behavior through controlled input changes. CheckList, contrast sets, and counterfactually augmented data expose local failures under small edits (Ribeiro et al., 2020; Gardner et al., 2020; Kaushik et al., 2020). Attribution methods instead use visual-region localization, attention flow, or image-token attention analysis (Selvaraju et al., 2017; Jain and Wallace, 2019; Abnar and Zuidema, 2020; Gong et al., 2024). HalluPrism follows the behavioral-testing view. It asks how the output changes under pressure on visual evidence, question-only priors, or cross-modal binding.

Mitigation and Routing. Hallucination mitigation often applies a fixed correction rule during decoding or after generation. Visual Contrastive Decoding, OPERA, M3ID, Woodpecker, and Volcano reduce hallucination through visual contrast,

decoding penalties, image-grounded amplification, or post-hoc revision (Leng et al., 2023; Huang et al., 2023; Favero et al., 2024; Yin et al., 2023; Lee et al., 2023). Recent inference-time methods refine this direction through inter-modality calibration, attention calibration, and visual-aware sparsification (Li et al., 2025; Fazli et al., 2025; Zhuang et al., 2025). HalluPrism is diagnosis-first. It profiles a risky prediction and then selects a matched prompt-based correction.

3 Methodology

We study multimodal uncertainty as a *pre-decision* diagnostic object. The goal is not to build a better scalar abstention score. We instead test whether a prediction carries a behavioral source signature that can diagnose failure family and route correction without assuming clean causal separability. The design therefore separates three objects that are often conflated: correctness ranking, failure-source diagnosis, and correction-path choice. This separation is important because the same source signature can be useful for explaining a failure, harmful when compressed into a scalar abstention score, and useful again when choosing a safer correction path. A diagnostic representation can therefore improve failure typing while still being the wrong object to compress into an answer-vs.-abstain score. All cited formal results appear in Appendix A.

Prediction and Source Signatures. Consider an image I where a traffic cone is to the right of a parked car, and a prompt q asks whether the cone is to the left of the car. If an MLLM M answers “yes” to $x = (I, q)$ while the reference answer y is “no”, the output is incorrect². We write correctness as $z(x, M) = 1[\text{norm}(\hat{y}) = \text{norm}(y)]$. Incorrect outputs receive a coarse failure-family label $f(x, M) \in \mathcal{F}$, such as object-hallucination, spatial/relation-error, attribute-error, or answerability-failure. A scalar confidence estimator maps the input-output instance to $c(x, M) \in [0, 1]$. For a tokenized answer $\hat{y} = (t_1, \dots, t_N)$, we use the length-normalized confidence probability:

$$c(\hat{y} \mid I, q) = \exp \left(\frac{1}{N} \sum_{i=1}^N \log p(t_i \mid t_{<i}, I, q) \right). \quad (1)$$

²We normalize answers before comparison, mapping surface variants such as “yes”, “Yes.”, or “yes, it is” to the same canonical task label.

For answerability-focused datasets, we use the dataset-specific normalized answer probability. This scalar score ranks likely correctness. However, it does not say whether the answer depends on weak visual evidence, survives without the image, or fails because the image-text relation is misbound. Hence, scalar confidence is a correctness-ranking surrogate, not a grounded reliability certificate. This non-identifiability is formalized in Proposition 1. We attach a $[0, 1]$ -bounded source signature to the same prediction:

$$s(x, M) = (V(x, M), L(x, M), A(x, M)). \quad (2)$$

Here V measures visual fragility, L measures language-prior reliance, and A measures cross-modal alignment instability. These scores are behavioral probe responses, not hidden causal mechanisms. They remain useful because failure families may occupy different regions of the joint (V, L, A) space.

Perturbation Probes. We measure each score by asking how the same prediction behaves under controlled input pressure. The clean answer $\hat{y}$ is the baseline. A probe changes one part of the input and produces $\tilde{y}$. We compare answers with $\text{flip}(\hat{y}, \tilde{y})$, which equals 1 when the answer changes, and confidence shifts with $\Delta c(\hat{y}, \tilde{y}) = |c(\hat{y}) - c(\tilde{y})|$. Flips capture discrete instability. Confidence shifts capture softer changes. The **visual score** V measures sensitivity to degraded visual evidence. We keep the prompt fixed and use Gaussian blur ($\sigma = 15$), center crop (50% per dimension, resized back), reduce brightness to 30% of original image, and Gaussian noise ($\mu = 0, \sigma = 25$). For the k -th visual perturbation answer $\tilde{y}_k^v$:

$$V = 0.6 \cdot \frac{1}{K} \sum_{k=1}^{K=4} \text{flip}(\hat{y}, \tilde{y}_k^v) + 0.4 \cdot \min \left(2 \cdot \frac{1}{K} \sum_{k=1}^{K=4} \Delta c(\hat{y}, \tilde{y}_k^v), 1 \right). \quad (3)$$

The larger flip weight makes changed answers the primary signal of visual fragility. The **language-prior score** L measures answer survival after removing image evidence. We replace the image with a uniform blank image I_{blank} and keep the prompt fixed. Let $\tilde{y}^\ell$ be the blank-image answer. We define:

$$L = \mathbf{1}[\text{match}(\hat{y}, \tilde{y}^\ell)] \cdot \min \left(\frac{c(\tilde{y}^\ell)}{c(\hat{y}) + \epsilon}, 1 \right). \quad (4)$$

A high L means that the same answer persists with similar confidence without the original image. It is a high-recall behavioral marker of answer survival, not a causal claim that the model ignores the image. Since Eq. 4 can saturate, we use the full (V, L, A) vector for diagnosis. Appendix D.1 reports match-only and denominator-stable controls. The **alignment score** A measures sensitivity to image-text binding pressure. We use grounding re-checking for all datasets and relation swapping when an explicit relation is available. VSR supports relation swaps because its captions contain relations such as *left*, *right*, *above*, and *behind*. In the cone-car example, a left-of question is swapped into the corresponding right-of question. A relation-sensitive model should not preserve the same normalized answer under both prompts. Let $\tilde{y}_{\text{ground}}^a$ be the grounding-prompt answer and $\tilde{y}_{\text{rel}}^a$ the relation-swapped answer when available. We define:

$$\begin{aligned} P_g &= 0.5 \cdot \text{flip}(\hat{y}, \tilde{y}_{\text{ground}}^a) \\ &\quad + \min(\Delta c(\hat{y}, \tilde{y}_{\text{ground}}^a), 0.5), \\ P_r &= \mathbf{1}[\text{available}] \cdot \mathbf{1}[\text{norm}(\tilde{y}_{\text{rel}}^a) = \text{norm}(\hat{y})], \\ A &= \begin{cases} \min((1 - \lambda)P_g + \lambda P_r, 1), & \text{if available,} \\ \min(P_g, 1), & \text{otherwise,} \end{cases} \\ \lambda &= 0.6. \end{aligned} \quad (5)$$

The relation component receives larger weight because it gives a sharper binding stress. The expected answer should change under an explicit relation swap. The grounding component gives a weaker universal alignment probe. We therefore avoid comparing raw A magnitudes across relation-bearing and non-relation-bearing datasets. Cross-dataset diagnosis uses dataset and model controls, while routing uses within-setting normalized dominance. Appendix D.2 reports the no-VSR control.

Interpreting And Using Source Signatures.

Once the probes produce (V, L, A) , we separate three questions. Which coordinate is largest? Does a targeted intervention mainly move its intended coordinate? Does the full signature help diagnosis, abstention, or routing? A large coordinate is not a causal assignment. Raw dominance only counts the largest coordinate, $s_{\text{raw}}^*(x, M) = \arg \max_{Q \in \{V, L, A\}} Q(x, M)$. Because one coordinate can saturate, we use within-setting normalized dominance for routing: $s_{\text{norm}}^*(x, M) = \arg \max_{Q \in \{V, L, A\}} z_Q(x, M)$, where $z_Q(x, M) =$

$(Q(x, M) - \mu_Q)/(\sigma_Q + \epsilon)$. This prevents a saturated coordinate, such as L , from dominating every routing decision. We next test whether the scores behave as separable labels. Let $\mathcal{T}_V$, $\mathcal{T}_L$, and $\mathcal{T}_A$ denote visual-ruined, language-ruined, and alignment-ruined conditions. These are stronger than the local probes used to compute the scores. For target R and coordinate $Q \in \{V, L, A\}$:

$$\Delta_{R,Q}(x, M) = Q(\mathcal{T}_R(x), M) - Q(x, M). \quad (6)$$

A cleanly separable signature would satisfy magnitude-based diagonal dominance:

$$|\Delta_{R,R}(x, M)| > \max_{Q \neq R} |\Delta_{R,Q}(x, M)|. \quad (7)$$

We test this condition rather than assume it. Definition A.1 states the condition. Proposition 2 formalizes the key limitation. If a source-targeted intervention changes another coordinate at least as much as the intended coordinate, the intervention cannot be read as source-specific. We then evaluate three uses of source signatures. Failure-family diagnosis asks whether $s(x, M)$ predicts $f(x, M)$ better than scalar confidence. The role separation follows from Proposition 3 and Theorem 1: a signature can help explain the kind of error without improving correctness ranking. Abstention scalarization asks whether a source-aware score $\tilde{c}(x, M) = g(c, V, L, A)$ improves answer-versus-abstain ranking. Proposition 4 and Corollary 1 state the ranking risk when source penalties are not aligned with correctness. Routing asks whether the signature helps choose a safer correction path. Proposition 5 and Corollary 2 motivate reporting fix rate and break rate with accuracy. For abstention, we test three scalarizations:

$$\begin{aligned} c_{\max} &= c \cdot (1 - \max(V, L, A)), \\ c_w &= c \cdot (1 - 0.3V - 0.3L - 0.4A), \\ c_{\text{geo}} &= c \cdot ((1 - V)(1 - L)(1 - A))^{1/3}. \end{aligned} \quad (8)$$

For intervention routing, the generic baseline applies the same reasoning prompt to every input. The matched policy maps $s_{\text{norm}}^*(x, M)$ to visual re-grounding, anti-prior prompting, or relation checking. For example, an alignment-dominant cone-car prediction is routed to relation checking rather than a generic reasoning prompt. We report accuracy, fix rate, and break rate:

$$\begin{aligned} \text{Fix} &= \frac{\#\{\text{incorrect} \rightarrow \text{correct}\}}{\#\{\text{incorrect}\}}, \\ \text{Break} &= \frac{\#\{\text{correct} \rightarrow \text{incorrect}\}}{\#\{\text{correct}\}}. \end{aligned} \quad (9)$$

The protocol uses one forward pass, four visual perturbation passes, one blank-image pass, and 1-2 alignment passes, depending on whether relation perturbation is available. No model is retrained.

4 Experimental Setup

We evaluate whether multimodal uncertainty should be studied as source-signature diagnosis and correction-path selection rather than only as answer-versus-abstain scoring. For each model output, we compute scalar confidence c and source signature $s(x) = (V, L, A)$. We test failure diagnosis, abstention, and intervention routing under matched perturbation protocols.

Datasets and Models. We evaluate Gemma 4, Qwen3-VL, LLaVA-Mistral, and LLaVA-Vicuna on HallusionBench, POPE, Visual Spatial Reasoning (VSR), and VizWiz-VQA (Google DeepMind, 2026; Bai et al., 2025; Liu et al., 2023b; Guan et al., 2024; Li et al., 2023; Liu et al., 2023a; Gurari et al., 2018). The suite covers visually grounded hallucination, object-presence hallucination, spatial relation binding, and answerability. We use the image-paired HallusionBench subset, all POPE splits, the VSR zero-shot development set, and the VizWiz-VQA validation split. We use greedy decoding with temperature 0, disabled sampling, and a 128-token cap. Split, filtering, checkpoint, and license details appear in Appendices B and G.

Diagnostic and Decision Tasks. Each prediction receives correctness and coarse failure-family labels. Correct predictions map to *no failure*. Incorrect predictions follow benchmark structure. POPE maps to object-presence failure, VSR to spatial or relation error, VizWiz-VQA to answerability failure, and HallusionBench to visually grounded hallucination categories. This tests broad failure-family separation rather than an exhaustive taxonomy. We train logistic regression, XGBoost, and LightGBM with stratified 5-fold cross-validation over $[c]$, $[V, L, A]$, and $[V, L, A, c]$ (Chen and Guestrin, 2016; Ke et al., 2017). We report diagnosis AUROC and Macro-F1, source profiles, abstention AUROC and risk-coverage, and routing accuracy, fix rate, and break rate. These metrics keep failure typing, answer-versus-abstain ranking, and correction effects separate. Appendix B.6 gives the full mapping.

Controls, Abstention, and Routing. Since failure families and datasets are partly aligned, we add dataset-ID and within-dataset controls. We

Dataset	N Per	Primary Stress	Source Probe Role
HallusionBench	951	Visual illusion, attribute error	Visual and alignment pressure
POPE	9,000	Object-presence hallucination	Blank-image and object-prior pressure
VSR	340	Spatial and relation binding	Relation-swap alignment pressure
VizWiz-VQA	4,319	Visual insufficiency, answerability	Visual degradation and blank-image pressure
Models	Gemma 4, LLaVA-Mistral, LLaVA-Vicuna, and Qwen3-VL under deterministic decoding.		

Table 1: Evaluation suite and roles. Evaluation roles focus on mixed failure-family diagnosis for HallusionBench, object hallucination/abstention for POPE, relation error/routing breakdown for VSR, and answerability diagnosis for VizWiz-VQA. Each model-dataset output receives scalar confidence c and source signature (V, L, A) . Full dataset roles, split details, probe availability, and failure-family mappings are given in Appendix B.

also test the two main score-design concerns. For L , Appendix D.1 compares the clipped-ratio score with match-only and denominator-stable variants. For A , Appendix D.2 removes VSR, the only relation-swap-eligible dataset. Abstention compares scalar confidence with Eq. 8. Routing compares no intervention, one generic reasoning prompt, and matched intervention. Matched routing selects image enhancement, anti-prior instructions, or grounding/relation checking from normalized source dominance. Exact templates and enhancement parameters appear in Appendix C.4. We summarize benchmark roles in Table 1.

Human Audit. We run a blind audit over 200 examples, with 50 from each dataset. Three annotators assign visual evidence insufficiency, language-prior reliance, alignment error, mixed, unclear, or no failure. Annotators do not see V , L , A , scalar confidence, or predicted source. We adjudicate 36 disagreement cases. Full-label Fleiss’ κ is 0.236, and source versus non-source Fleiss’ κ is 0.48. Appendix E gives the full protocol and tables.

5 Results

We evaluate (V, L, A) for failure typing and matched correction, not as a scalar confidence replacement. The supporting appendix evidence follows the same order: source profiles in Tables 12-13, entanglement in Tables 14-18, diagnosis in Tables 20-27, human audit in Tables 28-31, abstention in Tables 32-34, and routing in Tables 35-36.

Prevalence is Not Diagnostic Value. We observe from Table 2 that L is high across settings, with means from 0.91 to 1 and language-prior match from 37.6% to 96.5%. VizWiz gives the highest visual scores (0.19 to 0.30), while Gemma 4/VSR and Qwen3-VL/VSR give the highest alignment scores (0.68 and 0.61). Raw dominance is almost entirely language-prior. Overall, 58,287/58,440

samples are L -dominant, versus 151 for A and 2 for V . This is why we treat L as a high-recall answer-survival marker and use the full vector, not raw L dominance, for diagnosis. Appendix Tables 12-13 give the full source profiles and raw dominance counts. Appendix Table 27 gives matched source-stress slices that support the same reading.

Interventions Expose Entanglement. We test whether source-targeted ruined inputs isolate their intended coordinate under Definition A.1. We observe from Table 2 that only 18/48 checks pass magnitude-based diagonal dominance: visual 11/16, language 2/16, and alignment 5/16. Qwen3-VL/HallusionBench gives the main intuition. Visual ruin decreases V by 0.09 with small L/A changes, while language ruin decreases L by 0.09 and increases A by 0.18. Appendix Table 14 gives the pass counts, and Appendix Tables 15-18 give the model-specific signed deltas. These failures match Proposition 2. Hence, (V, L, A) is a joint source signature rather than three causal labels. This is, however, not a failure of the probe design. It is the diagnostic setting we want to expose: a visually fragile answer can also be language-prior sensitive or alignment-sensitive.

Signatures Diagnose Failure Families. Although the scores are entangled, their joint pattern remains diagnostic. Since failure-family labels and benchmark identity are partly aligned, we first test dataset and model controls. We observe from Table 3 that adding source features beyond these identifiers improves raw-family Macro-F1 from 0.71 to 0.78, and the stricter outcome-family setting from 0.31 to 0.35. Appendix Tables 23-24 give these controls. Fixed-dataset results are strongest on HallusionBench and VizWiz, weaker on POPE, and negative on VSR, as reported in Appendix Table 25. These results bound the claim: source signatures are most useful when multiple failure modes or

Evidence Block	Setting	Main Value
Mean language-prior score	All model-dataset settings	L ranges from 0.91 to 1
Blank-image match	All model-dataset settings	LP match ranges from 37.6% to 96.5%
Highest visual fragility	VizWiz settings	V ranges from 0.19 to 0.30
Highest alignment instability	Gemma 4/VSR, Qwen3-VL/VSR	$A = 0.68$ and $A = 0.61$
Raw dominant source	58,440 pooled samples	$L/A/V = 58,287/151/2$
Magnitude-based diagonal pass count	All intervention checks	18/48 pass
Visual-ruined target	Across 16 settings	11/16 pass
Language-ruined target	Across 16 settings	2/16 pass
Alignment-ruined target	Across 16 settings	5/16 pass
Qwen3-VL/HallusionBench	Visual-ruined deltas	$\Delta V = -0.09, \Delta L = +0.01, \Delta A = -0.01$
Qwen3-VL/HallusionBench	Language-ruined deltas	$\Delta V = -0.05, \Delta L = -0.09, \Delta A = +0.18$

Table 2: **Source-profile and Entanglement Summary.** LP match is exact-answer agreement under blank-image replacement. Diagonal pass counts test whether the intended source has the largest absolute response under the corresponding ruined condition. Full profiles and intervention deltas are given in Appendix Tables 12-18.

Failure-Family Diagnosis				Tree-Based Feature Importance			
Classifier	Scalar	$V + L + A$	$V + L + A + c$	Feature	XGB	LGBM	Role
Logistic Regression	0.75	0.83	0.87	A	0.50	0.36	Alignment
XGBoost	0.78	0.95	0.97	L	0.20	0.15	Language prior
LightGBM	0.78	0.95	0.97	c	0.15	0.27	Confidence
				V	0.15	0.22	Visual fragility

Dataset-ID Controls				
Target	Baseline	Source Add-On	Δ AUROC	Δ Macro-F1
Raw family	$D + M + c$	$+ V + L + A$	+0.005	+0.078
Outcome family	$D + M + c$	$+ V + L + A$	+0.006	+0.040

Table 3: Failure-family diagnosis, feature importance, and dataset-ID controls. Classifier rows report pooled weighted one-vs-rest AUROC over 58,440 examples. Feature rows report pooled tree-based feature importance. Dataset-ID controls use LightGBM and report gains after adding source features beyond dataset identity D , model identity M , and scalar confidence c . Full-precision diagnosis and control results are given in Appendix Tables 21-25.

Control	Setting	AUROC	Macro-F1
L_{default}	clipped ratio	0.828	0.333
L_{match}	match only	0.823	0.346
L_{stable}	floored denominator	0.823	0.349
c	excluding VSR	0.761	0.329
(V, L, A)	excluding VSR	0.839	0.422

Table 4: Probe-design controls. The language-prior rows replace only the L coordinate inside the pooled (V, L, A) diagnosis feature set. The no-VSR control removes the only relation-swap-eligible dataset and compares scalar confidence with (V, L, A) under the same non-VSR scope. Details are in Appendix Tables 10-11

answerability-sensitive distinctions remain, not as a uniformly transferable rule across all benchmarks.

We also test whether the result is carried by probe-design choices. We observe from Table 4 that replacing the clipped-ratio L with match-only or denominator-stable variants keeps diagnosis stable. Removing VSR, the only relation-swap-eligible dataset, also preserves the gain of (V, L, A)

over scalar confidence. Thus, the result is not explained by the clipped-ratio form of L or by the VSR-only relation-swap branch. In the main pooled tree-based setting, XGBoost improves from 0.78 AUROC with scalar confidence to 0.95 with (V, L, A) , and to 0.97 with confidence added. Appendix Table 21 gives the full diagnosis results. The gain is not merely prevalence-driven: the alignment coordinate contributes 0.50 of pooled XGBoost importance, as detailed in Appendix Table 22. Appendix Table 20 reports source means by failure family. A blind human audit gives bounded semantic support: clear-source labels are mostly alignment (71/107), and (V, L, A) improves clear-source AUROC over scalar confidence from 0.43 to 0.60. Appendix Tables 28-31 give the audit protocol, agreement, confusion matrix, and audit-side classifier.

Scalarization is The Wrong Target. The diagnostic gain does not mean that source signatures should be collapsed into a better abstention score.

Abstention: Scalar Confidence Vs. Best Source-Aware Scalarization				
Model Family	Metric	Scalar Confidence	Best Source-Aware	
Gemma 4	AUROC	0.81	0.77 (0.04 ↓)	
LLaVA-Mistral	AUROC	0.84	0.81 (0.03 ↓)	
LLaVA-Vicuna	AUROC	0.83	0.78 (0.05 ↓)	
Qwen3-VL	AUROC	0.81	0.73 (0.08 ↓)	

Routing: No Intervention Vs. Generic Prompting Vs. Matched Routing				
Setting	Metric	No Intervention	Generic	Matched
Pooled	Accuracy	76.6%	75.7%	75.9%
Highest accuracy	Count	6/16	3/16	7/16
Generic vs. matched	Accuracy wins	–	6/16	10/16
Generic vs. matched	Lower break rate	–	6/16	10/16
Qwen3-VL/VSR	Accuracy	87.4%	76.2%	87.6%
Qwen3-VL/VSR	Break	–	15.5%	2.7%

Table 5: **Decision-facing behavior.** Abstention compares scalar confidence against the best source-aware scalarization for each model family. Routing compares no intervention, generic prompting, and matched routing. The *Highest accuracy* row counts which policy obtains the best accuracy in each model-dataset setting. The pairwise rows compare generic prompting and matched routing across the same 16 settings. Full-precision results appear in Appendix Tables 32-36.

We observe from Table 5 that weighted, geometric, and max-penalty scalarizations underperform in AUROC. Qwen3-VL drops from 0.81 scalar AUROC to 0.73 for the best source-aware score. Appendix Table 32 gives aggregate scalarization results, Appendix Table 33 gives per-dataset comparisons, and Appendix Table 34 gives risk-coverage values. This does not certify scalar confidence as grounded. Rather, it shows that a diagnostic feature can be useful for explaining the kind of failure while still being misaligned with answer-versus-abstain ranking. In other words, source-aware uncertainty should not be naively scalarized for abstention. As formalized in Appendix Proposition 4 and Corollary 1, source penalties and near-saturated coordinates can reverse useful confidence orderings.

Diagnosis-Triggered Routing. We test whether source signatures help choose safer correction paths. Generic prompting applies one reasoning prompt to every example, while matched routing selects visual re-grounding, anti-prior prompting, or relation checking from normalized source dominance. We observe from Table 5 that matched routing is not a universal accuracy booster. It beats generic prompting in 10/16 settings, with pooled accuracy 75.9% versus 75.7% for generic prompting and 76.6% without intervention. Its value is clearer in break-rate control. On Qwen3-VL/VSR, matched routing avoids the generic accuracy drop from 87.4% to 76.2% and lowers breakage from 15.5% to 2.7%, while maintaining 87.6% accu-

racy. It also lowers breakage on Gemma 4/HallusionBench and LLaVA-Mistral/VizWiz (15.1% → 4.9% and 21.4% → 17.4%). Appendix Table 35 gives the full routing results, and Appendix Table 36 gives model-level win counts. This supports routing when generic correction risks damaging correct predictions. The practical use is therefore not universal correction, but avoiding avoidable damage when a generic correction prompt is too blunt. Since baseline accuracy exceeds 0.5, Corollary 2 gives larger weight to avoiding broken correct answers than to an equal-sized repair gain.

6 Conclusion

In this paper, we present HalluPrism as a source-signature diagnostic for multimodal uncertainty in MLLMs. Across 58K+ examples, source signatures improve failure-family diagnosis, while matched routing reduces harmful breakage in selected high-risk settings. The same evidence shows a boundary: better failure typing does not imply better abstention scoring. We argue that the pursuit of a universally reliable scalar abstention verdict is ill-posed under open-world deployment since the validity of such guardrails depends on deployment-transfer assumptions (formalized in Appendix Theorem 2 and Proposition 6). Source-aware uncertainty should therefore reveal whether risk is visually fragile, language-prior-driven, or alignment-sensitive, and help determine whether a generic correction is safe.

Limitations

HalluPrism is a behavioral diagnostic layer, not a causal explanation of hidden mechanisms inside a Multimodal Large Language Model (MLLM). Its scores depend on the probe family, so (V, L, A) is a source signature rather than independent ground-truth causes. The evaluation covers four public benchmarks and four open-weight MLLM families, not video, multi-image reasoning, medical, or document-heavy expert settings, or latency-critical deployments. The implementation requires 7 to 8 forward passes per example. Routing uses lightweight prompts and gives conditional actionability, not universal mitigation. The human audit is intentionally small and checks semantic plausibility, not causal source recovery. The no-free-lunch argument does not reject selective prediction, calibration, risk control, or conformal prediction for their stated statistical targets. It states that these guarantees require validation-to-deployment assumptions and, by themselves, do not provide instance-grounded reliability.

Ethics Statement

HalluPrism is intended for model auditing and diagnostic analysis, not for standalone safety certification or deployment approval. Its scores are behavioral source signatures from controlled probes and should not be treated as causal explanations of hidden model mechanisms. The main ethical risk is over-trust, where a compact source label is mistaken for a complete reliability guarantee. We therefore keep diagnosis, scalarization, fix rate, and break rate separate. The experiments use public multimodal benchmarks and publicly accessible model families. We do not collect new user images or train a deployed model. Since VizWiz-VQA contains images and questions from blind or low-vision users, any release should avoid raw-image redistribution unless allowed by the original license. The human audit uses benchmark examples only, and annotators do not see model-derived source scores or routing decisions.

References

Samira Abnar and Willem Zuidema. 2020. Quantifying attention flow in transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics.

- Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, et al. 2025. [Qwen3-VL technical report](#).
- Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. Association for Computing Machinery.
- C. K. Chow. 1970. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46.
- Julian Martin Eisenschlos, Hernán Maina, Guido Ivetta, and Luciana Benotti. 2024. [Selectively answering visual questions](#).
- Alessandro Favero, Luca Zancato, Matthew Trager, Sidharth Choudhary, Pramuditha Perera, Alessandro Achille, Ashwin Swaminathan, and Stefano Soatto. 2024. [Multi-modal hallucination control by visual information grounding](#).
- Mehrdad Fazli, Bowen Wei, and Ziwei Zhu. 2025. [Mitigating hallucination in large vision-language models via adaptive attention calibration](#).
- Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, et al. 2023. [MME: A comprehensive evaluation benchmark for multimodal large language models](#).
- Matt Gardner, Yoav Artzi, Victoria Basmova, Jonathan Berant, Ben Bogin, et al. 2020. Evaluating models’ local decision boundaries via contrast sets. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1307–1323. Association for Computational Linguistics.
- Yonatan Geifman and Ran El-Yaniv. 2017. [Selective classification for deep neural networks](#).
- Yonatan Geifman and Ran El-Yaniv. 2019. SelectiveNet: A deep neural network with an integrated reject option. In *Proceedings of the 36th International Conference on Machine Learning*, pages 2151–2159. PMLR.
- Jiahui Geng, Fengyu Cai, Yuxia Wang, Heinz Koepl, Preslav Nakov, and Iryna Gurevych. 2023. [A survey of confidence estimation and calibration in large language models](#).
- Xuan Gong, Tianshi Ming, Xinpeng Wang, and Zhihua Wei. 2024. [DAMRO: Dive into the attention mechanism of LVLM to reduce object hallucination](#).
- Google DeepMind. 2026. [Gemma 4 Model Card](#). Accessed: 2026-05-20.
- Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.

693	Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian,	Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023.	750
694	Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen,	Semantic uncertainty: Linguistic invariances for un-	751
695	Furong Huang, Yaser Yacoob, Dinesh Manocha, and	certainty estimation in natural language generation.	752
696	Tianyi Zhou. 2024. Hallusionbench: An advanced		
697	diagnostic suite for entangled language hallucination	Seongyun Lee, Sue Hyun Park, Yongrae Jo, and Min-	753
698	and visual illusion in large vision-language models.	joon Seo. 2023. Volcano: Mitigating multimodal	754
699	In <i>Proceedings of the IEEE/CVF Conference on Com-</i>	hallucination through self-feedback guided revision.	755
700	<i>puter Vision and Pattern Recognition</i> , pages 14375–		
701	14385.		
702	Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Wein-	Sicong Leng, Hang Zhang, Guanzheng Chen, Xin	756
703	berger. 2017. On calibration of modern neural net-	Li, Shijian Lu, Chunyan Miao, and Lidong Bing.	757
704	works. In <i>Proceedings of the 34th International</i>	2023. Mitigating object hallucinations in large vision-	758
705	<i>Conference on Machine Learning</i> , pages 1321–1330.	language models through visual contrastive decoding.	759
706	PMLR.		
707	Danna Gurari, Qing Li, Abigale J. Stangl, Anhong Guo,	Jiaming Li, Jiacheng Zhang, Zequn Jie, Lin Ma, and	760
708	Chi Lin, Kristen Grauman, Jiebo Luo, and Jeffrey P.	Guanbin Li. 2025. Mitigating hallucination for large	761
709	Bigham. 2018. Vizwiz grand challenge: Answering	vision language model by inter-modality correlation	762
710	visual questions from blind people. In <i>Proceedings</i>	calibration decoding.	763
711	<i>of the IEEE Conference on Computer Vision and</i>		
712	<i>Pattern Recognition</i> , pages 3608–3617.	Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang,	764
713	Qidong Huang, Xiaoyi Dong, Pan Zhang, Bin Wang,	Wayne Xin Zhao, and Ji-Rong Wen. 2023. Eval-	765
714	Conghui He, Jiaqi Wang, Dahua Lin, Weiming	uating object hallucination in large vision-language	766
715	Zhang, and Nenghai Yu. 2023. OPERA: Alleviating	models. In <i>Proceedings of the 2023 Conference on</i>	767
716	hallucination in multi-modal large language models	<i>Empirical Methods in Natural Language Process-</i>	768
717	via over-trust penalty and retrospection-allocation.	<i>ing</i> , pages 292–305. Association for Computational	769
718	Tao Huang, Zhekun Liu, Rui Wang, Yang Zhang, and	Linguistics.	770
719	Liping Jing. 2025. Visual hallucination detection in	Fangyu Liu, Guy Emerson, and Nigel Collier. 2023a.	771
720	large vision-language models via evidential conflict.	Visual spatial reasoning. <i>Transactions of the Associ-</i>	772
721	Sarthak Jain and Byron C. Wallace. 2019. Attention	<i>ation for Computational Linguistics</i> , 11:635–651.	773
722	is not explanation. In <i>Proceedings of the 2019 Con-</i>		
723	<i>ference of the North American Chapter of the Asso-</i>	Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae	774
724	<i>ciation for Computational Linguistics: Human Lan-</i>	Lee. 2023b. Improved baselines with visual instruc-	775
725	<i>guage Technologies</i> . Association for Computational	tion tuning.	776
726	Linguistics.		
727	Saurav Kadavath, Tom Conerly, Amanda Askell, Tom	Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li,	777
728	Henighan, Dawn Drain, Ethan Perez, Nicholas	Songyang Zhang, et al. 2023c. MMBench: Is your	778
729	Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli	multi-modal model an all-around player?	779
730	Tran-Johnson, Scott Johnston, Sheer El-Showk,		
731	Andy Jones, Nelson Elhage, Tristan Hume, Anna	Potsawee Manakul, Adian Liusie, and Mark J. F. Gales.	780
732	Chen, Yuntao Bai, Sam Bowman, Stanislav Fort,	2023. SelfCheckGPT: Zero-resource black-box hal-	781
733	Deep Ganguli, Danny Hernandez, Josh Jacobson,	lucination detection for generative large language	782
734	Jackson Kernion, Shauna Kravec, Liane Lovitt, Ka-	models.	783
735	mal Ndousse, Catherine Olsson, Sam Ringer, Dario	Erum Mushtaq, Zalan Fabian, Yavuz Faruk Bak-	784
736	Amodei, Tom Brown, Jack Clark, Nicholas Joseph,	man, Anil Ramakrishna, Mahdi Soltanolkotabi, and	785
737	Ben Mann, Sam McCandlish, Chris Olah, and Jared	Salman Avestimehr. 2025. HARMONY: Hidden ac-	786
738	Kaplan. 2022. Language models (mostly) know what	tivity representations and model output-aware un-	787
739	they know.	certainty estimation for vision-language models.	788
740	Divyansh Kaushik, Eduard Hovy, and Zachary C. Lip-	Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin,	789
741	ton. 2020. Learning the difference that makes a dif-	and Sameer Singh. 2020. Beyond accuracy: Be-	790
742	ference with counterfactually-augmented data. In	havioral testing of NLP models with CheckList. In	791
743	<i>International Conference on Learning Representa-</i>	<i>Proceedings of the 58th Annual Meeting of the Asso-</i>	792
744	<i>tions.</i>	<i>ciation for Computational Linguistics</i> , pages 4902–	793
745	Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang,	4912. Association for Computational Linguistics.	794
746	Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu.		
747	2017. LightGBM: A highly efficient gradient boost-	Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns,	795
748	ing decision tree. In <i>Advances in Neural Information</i>	Trevor Darrell, and Kate Saenko. 2018. Object hallu-	796
749	<i>Processing Systems</i> , volume 30.	cination in image captioning. In <i>Proceedings of the</i>	797
		<i>2018 Conference on Empirical Methods in Natural</i>	798
		<i>Language Processing</i> . Association for Computational	799
		Linguistics.	800
		Ramprasaath R. Selvaraju, Michael Cogswell, Ab-	801
		hishek Das, Ramakrishna Vedantam, Devi Parikh,	802

and Dhruv Batra. 2017. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 618–626.

Hoigi Seo, Dong Un Kang, Hyunjin Cho, Joohoon Lee, and Se Young Chun. 2025. [On epistemic uncertainty of visual tokens for object hallucinations in large vision-language models.](#)

Tejas Srinivasan, Jack Hessel, Tanmay Gupta, Bill Yuchen Lin, Yejin Choi, Jesse Thomason, and Khyathi Raghavi Chandu. 2024. [Selective “selective prediction”: Reducing unnecessary abstention in vision-language reasoning.](#)

Junyang Wang, Yuhang Wang, Guohai Xu, Jing Zhang, Yukai Gu, Haitao Jia, Jiaqi Wang, Haiyang Xu, Ming Yan, Ji Zhang, and Jitao Sang. 2023. [AMBER: An LLM-free multi-dimensional benchmark for MLLMs hallucination evaluation.](#)

Shukang Yin, Chaoyou Fu, Sirui Zhao, Tong Xu, Hao Wang, Dianbo Sui, Yunhang Shen, Ke Li, Xing Sun, and Enhong Chen. 2023. [Woodpecker: Hallucination correction for multimodal large language models.](#)

Xianwei Zhuang, Zhihong Zhu, Yuxin Xie, Liming Liang, and Yuexian Zou. 2025. [VASparse: Towards efficient visual hallucination mitigation for large vision-language model via visual-aware sparsification.](#)

A Theoretical Analysis

We provide formal results that support the source-signature interpretation used in the main paper. The results do not claim that V , L , and A recover hidden causal mechanisms inside a Multimodal Large Language Model (MLLM). They instead separate four roles: scalar confidence ranks correctness, source signatures support failure-family diagnosis, scalarization can distort abstention rankings, and routing must be evaluated through both fix and break rates. The setup is related to classical reject-option classification and selective prediction (Chow, 1970; Geifman and El-Yaniv, 2017, 2019).

A.1 Scalar Confidence Does Not Identify Failure Source

Let X denote the input-output instance, $C = c(X) \in [0, 1]$ denote scalar confidence, $Z \in \{0, 1\}$ denote correctness, and $F \in \mathcal{F}$ denote the failure-family label, with $F = \text{None}$ when $Z = 1$. A source-diagnosis rule based only on scalar confidence is any measurable map $h : [0, 1] \rightarrow \mathcal{F} \cup \{\text{None}\}$.

Proposition 1 (Scalar Non-Identifiability of Failure Source). *A zero-error diagnosis rule $h(C)$ exists*

if and only if F is measurable with respect to C . Equivalently, for almost every confidence value c , the conditional distribution $P(F \mid C = c)$ must place all mass on one label. If there exists a set $B \subseteq [0, 1]$ with $P(C \in B) > 0$ such that

$$\mathbb{E} \left[1 - \max_{f \in \mathcal{F} \cup \{\text{None}\}} P(F = f \mid C) \mid C \in B \right] > 0, \quad (10)$$

then every scalar-confidence-only diagnosis rule has positive error on B .

Proof. If F is measurable with respect to C , then there exists a measurable function h such that $F = h(C)$ almost surely. This gives zero diagnosis error. Conversely, if a zero-error rule $h(C)$ exists, then $F = h(C)$ almost surely, and hence F is measurable with respect to C . For the positive-error statement, condition on each confidence value. The best scalar rule at confidence value $C = c$ selects the label with largest conditional probability, so its conditional error is $1 - \max_f P(F = f \mid C = c)$. Averaging this quantity over $C \in B$ gives the displayed expression. If this value is positive, every scalar-confidence-only diagnosis rule has positive error on B . $\square$

Implications. Proposition 1 explains why scalar confidence is an incomplete object for multimodal hallucination analysis. Even if scalar confidence is useful for abstention, it cannot diagnose whether an error is an object hallucination, a spatial error, an attribute error, or an answerability failure unless those labels are already determined by confidence. This motivates the failure-family diagnosis experiment, where (V, L, A) is evaluated against scalar confidence for predicting the type of failure.

A.2 Interventional Separability and Entanglement

Let $S(X) = (V(X), L(X), A(X))$ denote the source signature. Let $\mathcal{T}_V$, $\mathcal{T}_L$, and $\mathcal{T}_A$ denote visual, language-prior, and alignment interventions. For $R, S \in \{V, L, A\}$, define the intervention response:

$$\Delta_{R,S}(X) = S(\mathcal{T}_R(X)) - S(X). \quad (11)$$

The expected response matrix is $D \in \mathbb{R}^{3 \times 3}$, with entries:

$$D_{R,S} = \mathbb{E}[\Delta_{R,S}(X)]. \quad (12)$$

Definition A.1 (Diagonal Separability Under A Probe Family). The source signature is diagonally separable under the probe family $\{\mathcal{T}_V, \mathcal{T}_L, \mathcal{T}_A\}$ if, for every $R \in \{V, L, A\}$,

$$D_{R,R} > \max_{S \neq R} D_{R,S}. \quad (13)$$

Proposition 2 (Off-Diagonal Magnitude Response Implies Probe-Level Entanglement). *If Definition A.1 fails for some source-targeting ruined condition R , then the probe family does not support a one-to-one interpretation in which condition R primarily affects only source R . In particular, if $|D_{R,S}| \geq |D_{R,R}|$ for some $S \neq R$, then the observed response to $\mathcal{T}_R$ is at least as large in magnitude on an off-target source as on its intended source.*

Proof. The condition for a one-to-one source-selective probe interpretation is magnitude-based diagonal dominance under Definition A.1. If $|D_{R,S}| \geq |D_{R,R}|$ for some $S \neq R$, then the off-target response has magnitude no smaller than the intended response. Hence the intervention response cannot be assigned uniquely to the intended source under this probe family. $\square$

Implications. Proposition 2 gives the formal basis for the interventional entanglement matrix. The empirical question is not whether every ruined condition validates a clean source label. The question is whether each source-targeting condition produces the largest absolute response on its intended coordinate or whether comparable off-target responses appear. When a visual-ruined condition changes alignment or language-prior behavior, that response is informative rather than a nuisance. It supports the interpretation of (V, L, A) as a joint source signature.

A.3 Diagnosis And Abstention Are Different Prediction Problems

Let T denote the failure type conditional on being wrong, so that $T = F$ when $Z = 0$. Let C denote scalar confidence and let $\mathbf{S} = (V, L, A)$ denote the source signature. Abstention requires ranking examples by $P(Z = 1 \mid \cdot)$. Failure diagnosis requires predicting T or F .

Proposition 3 (Diagnostic Information Need Not Improve Correctness Ranking). *There exist distributions over $(C, \mathbf{S}, Z, T)$ such that the source signature $\mathbf{S}$ is strictly more informative than scalar*

confidence C for failure-type prediction among incorrect examples, while C is at least as informative as $\mathbf{S}$ for correctness ranking. Here $C \in [0, 1]$ denotes scalar confidence, $\mathbf{S} = (V, L, A)$ denotes the source signature, $Z \in \{0, 1\}$ denotes correctness, and T denotes the failure type conditional on $Z = 0$. One sufficient condition is:

$$\begin{aligned} T &\not\perp \mathbf{S} \mid Z = 0, \\ T &\perp C \mid Z = 0, \\ P(Z = 1 \mid C, \mathbf{S}) &= P(Z = 1 \mid C). \end{aligned} \quad (14)$$

Under these conditions, $\mathbf{S}$ can improve failure-type diagnosis over C among incorrect examples, although adding $\mathbf{S}$ cannot improve the Bayes-optimal correctness score beyond C .

Proof. The first two conditions state that S carries information about the failure type among incorrect examples, while C does not. Therefore a classifier with access to S can improve failure-type prediction over a classifier using C alone whenever the conditional distributions differ across failure types. The third condition states that, after conditioning on C , the source signature provides no additional information about correctness. Hence the Bayes-optimal correctness ranking is a function of C alone. Adding S cannot improve that ranking. $\square$

Implications. Proposition 3 explains why the failure-diagnosis and abstention results can move in opposite directions. The source signature can substantially improve failure-family AUROC while still failing to improve correctness ranking. This supports the paper’s central distinction. Source signatures are diagnostic representations. They are not guaranteed scalar confidence improvements.

Theorem 1 (Role Separation Between Source Signatures And Scalar Confidence). *Suppose the conditions in Proposition 3 hold. Then any Bayes-optimal abstention policy based on (C, S) can be matched by a policy based on C alone, while a Bayes-optimal failure-family classifier based on (C, S) can strictly outperform every classifier based on C alone.*

Proof. For abstention, the Bayes-optimal score is $P(Z = 1 \mid C, S)$. By the third condition in Eq. 14, this equals $P(Z = 1 \mid C)$. Therefore a policy that uses only C can match the optimal policy that uses (C, S) . For failure-family prediction, the first two conditions imply that S changes the conditional

distribution of T among incorrect examples, while C does not. Therefore the Bayes-optimal classifier with access to S can achieve lower expected classification error than any classifier restricted to C when the conditional failure-type distributions differ across source-signature regions. $\square$

Implications. Theorem 1 states that a source signature may be valuable for explaining what kind of failure occurred without improving answer-versus-abstain decisions. This justifies evaluating failure-family diagnosis and abstention as separate tasks, rather than treating abstention as the only test of uncertainty usefulness.

A.4 Why Naive Source-Aware Scalarization Can Hurt Abstention

Many source-aware confidence rules have the multiplicative form:

$$\tilde{C} = C \cdot \phi(S), \quad (15)$$

where $\phi(S) \in [0, 1]$ is a source penalty. The weighted, geometric, and max-penalty scores in Section 3 are instances of this pattern.

Proposition 4 (Pairwise Inversion Under Source Penalties). *Consider two examples i and j with scalar confidences $C_i > C_j$. A multiplicative source-aware score $\tilde{C} = C\phi(S)$ reverses their ranking if and only if:*

$$\frac{\phi(S_j)}{\phi(S_i)} > \frac{C_i}{C_j}. \quad (16)$$

Therefore, source penalties can hurt abstention whenever penalty variation is not aligned with correctness and is large enough to overcome scalar-confidence margins.

Proof. The source-aware ranking reverses the scalar-confidence ranking when $\tilde{C}_j > \tilde{C}_i$. This is equivalent to $C_j\phi(S_j) > C_i\phi(S_i)$. Dividing both sides by $C_j\phi(S_i)$ gives Eq. 16. $\square$

Implications. Proposition 4 explains how source-aware scalarization can reduce abstention quality. A penalty based on V , L , and A can reorder examples that scalar confidence ranks correctly. This matters especially when the penalty is weakly related to correctness. The result supports the source-aware confidence-trap analysis in Section 5.

Corollary 1 (Saturated Source Scores Can Compress Useful Rankings). *Suppose a source penalty*

$\phi(S)$ is strongly affected by a nearly saturated coordinate such as $L \approx 1$. If $\phi(S)$ varies across examples due to small changes in saturated source scores and those variations are weakly correlated with correctness, then the multiplicative score $\tilde{C} = C\phi(S)$ can introduce pairwise inversions relative to C without adding useful correctness information.

Proof. By Proposition 4, ranking inversions occur whenever the penalty ratio exceeds the scalar-confidence ratio. Near saturation, small absolute variations in a penalty function can still produce relative penalty differences across examples, especially for nonlinear or max-based penalties. If those penalty differences are weakly correlated with correctness, they can reverse confidence orderings without improving correctness ranking. Therefore the scalarized score can compress or distort useful confidence rankings. $\square$

Implications. Corollary 1 connects the formal ranking argument to the empirical source profiles. The language-prior score is near saturation in many model-dataset settings. Multiplicative source penalties therefore risk flattening or distorting the confidence ranking. This explains why source signatures can improve failure diagnosis while source-aware scalar confidence can underperform scalar confidence in selective prediction.

A.5 Routing Depends On Fix And Break Rates

Let a be the baseline accuracy before intervention. For a policy π , let Fix_π be the fraction of originally incorrect examples corrected by the intervention and let Break_π be the fraction of originally correct examples made incorrect.

Proposition 5 (Accuracy Decomposition For Intervention Routing). *The post-intervention accuracy of policy π is:*

$$\text{Acc}_\pi = a + (1 - a)\text{Fix}_\pi - a\text{Break}_\pi. \quad (17)$$

For two policies π_1 and π_2 , policy π_1 has higher post-intervention accuracy than π_2 if and only if:

$$(1 - a)(\text{Fix}_{\pi_1} - \text{Fix}_{\pi_2}) > a(\text{Break}_{\pi_1} - \text{Break}_{\pi_2}). \quad (18)$$

Proof. Among the originally correct fraction a , the intervention preserves a fraction $1 - \text{Break}_\pi$.

Among the originally incorrect fraction $1 - a$, it repairs a fraction Fix_π . Thus:

$$\begin{aligned} \text{Acc}_\pi &= a(1 - \text{Break}_\pi) + (1 - a)\text{Fix}_\pi, \\ \text{Acc}_\pi &= a + (1 - a)\text{Fix}_\pi - a\text{Break}_\pi. \end{aligned} \quad (19)$$

Subtracting the expression for π_2 from the expression for π_1 gives Equation. 18. $\square$

Implications. Proposition 5 explains why intervention routing should be evaluated with both fix rate and break rate. A generic intervention can repair some failures and still reduce accuracy if it breaks many originally correct examples. This is especially important in high-accuracy settings, where the $a\text{Break}_\pi$ term is large. The result supports the paper’s routing analysis, where matched routing is most meaningful when it avoids harmful generic overcorrection.

Corollary 2 (Break-Rate Reduction Can Matter More Than Fix-Rate Gain). *If baseline accuracy $a > 0.5$, then a one-point reduction in break rate contributes more to post-intervention accuracy than a one-point increase in fix rate. Formally, the coefficient on Break_π in Eq. 17 has magnitude a , while the coefficient on Fix_π is $1 - a$.*

Proof. The claim follows directly from Eq. 17. The contribution of fix rate is weighted by $1 - a$, while the contribution of break rate is weighted by a . If $a > 0.5$, then $a > 1 - a$, and break-rate changes have larger absolute effect on final accuracy than equal-sized fix-rate changes. $\square$

Implications. Corollary 2 clarifies why the break rate is emphasized alongside accuracy. In settings such as Qwen3-VL/VSR, the generic intervention harms many originally correct answers. Reducing breakage can be more safety-relevant than maximizing repair count alone. This supports the conditional routing claim: source signatures are useful when they prevent harmful interventions, not only when they increase pooled accuracy.

A.6 Limits Of Scalar Abstention Under Open-World Deployment

The preceding results imply a broader limitation of score-based abstention. Abstention is not an intrinsic inference-time capability of a model. It is a designer-imposed guardrail over a score, threshold, prediction set, verifier, judge, or learned rejection rule. Such a guardrail can support selective answering under validation assumptions. However, it

is not automatically an instance-grounded verdict for a new input. We separate two requirements. First, the scalar score must be sufficient for the grounded safe-answer label. Second, the validation-to-deployment score-risk relation must transfer to the inference distribution. Without these requirements, a score-based abstention rule can be executed, but it has no unconditional inference-time guarantee.

Theorem 2 (No Grounded Abstention From A Non-Sufficient Scalar). *Let $X = (I, q, \hat{y})$ be a prediction instance. Let $Y^* \in \{0, 1\}$ denote the grounded safe-answer label, where $Y^* = 1$ means that the model should answer under the target reliability criterion, and $Y^* = 0$ means that it should not answer. Let $R = r(X) \in \mathbb{R}$ be any scalar abstention score, and let an abstention policy be any measurable function $h(R) \in \{0, 1\}$.*

A perfectly reliable scalar abstention policy exists if and only if Y^ is measurable with respect to R . Equivalently, for almost every score value or score region used by the policy, all examples assigned the same scalar information must have the same grounded safe-answer label. If there exists a score region B with $P(R \in B) > 0$ such that*

$$\begin{aligned} P(Y^* = 1 \mid R \in B) &> 0, \\ P(Y^* = 0 \mid R \in B) &> 0, \end{aligned} \quad (20)$$

then every abstention policy based only on R has positive error on B . Therefore, unless the scalar score is sufficient for the grounded safe-answer label, the question “should the model answer?” cannot be reliably answered from that scalar alone.

Proof. If Y^* is measurable with respect to R , then there exists a measurable function h such that $h(R) = Y^*$ almost surely. This gives a perfectly reliable scalar abstention policy. Conversely, suppose a perfectly reliable scalar abstention policy $h(R)$ exists. Then $Y^* = h(R)$ almost surely, which means Y^* is measurable with respect to R . For the positive-error statement, condition on $R \in B$. Any policy based only on R must assign its decision using only the scalar information available in B . Since both $Y^* = 1$ and $Y^* = 0$ have positive conditional mass in B , the policy must be wrong on the mass assigned to at least one of these labels. Hence, the error on B is positive. $\square$

Implication. The theorem is score-agnostic. It applies to confidence, entropy, calibrated probability, semantic uncertainty, verifier scores, judge

scores, conformal nonconformity scores, and learned rejection scores. A scalar score can support reliable abstention only when it already contains the grounded safe-answer state required by the deployment criterion.

Proposition 6 (No Unconditional Inference-Time Abstention Guarantee). *Let $R = r(X)$ be any scalar score used for abstention, and let τ be a threshold chosen from validation data. The inference-time decision*

$$\text{answer}(X) = \mathbf{1}[R(X) \geq \tau] \quad (21)$$

is guaranteed to satisfy a target deployment risk only under a transfer condition connecting the validation and deployment distributions, for example

$$\begin{aligned} P_{\text{deploy}}(Y^* = 0 \mid R \geq \tau) \\ = P_{\text{val}}(Y^* = 0 \mid R \geq \tau). \end{aligned} \quad (22)$$

Without such a condition, there exists a deployment distribution with the same score range and the same abstention rule for which the accepted region has arbitrarily high error. Hence, score-based abstention has no distribution-free inference-time guarantee over open-world inputs.

Proof. The threshold τ is fixed after validation. If no assumption links P_{deploy} to P_{val} , we can construct a deployment distribution that places positive, or even all, mass on examples with $R \geq \tau$ and $Y^* = 0$. The abstention rule accepts these examples because it observes only R . Therefore, the accepted risk on deployment can be arbitrarily larger than the validation risk. Any guarantee must therefore come from an external transfer assumption, not from the score-threshold rule itself. $\square$

Implication. The proposition locates the open-world weakness of score-threshold abstention. A validation risk curve, calibration curve, or prediction-set guarantee depends on the validation support and the assumed transfer relation. The accepted region is therefore warranted by the designer’s validation assumptions, not by an unconditional inference-time guarantee for each new instance.

Corollary 3 (No Source-Complete Scalar Reliability). *Let G denote a decision-relevant grounding or failure-source state, such as whether the answer is visually supported, language-prior driven, or cross-modally bound. Let A^* denote the safe correction action, with $A^* = \text{NONE}$ when no correction is*

needed. If either G or A^ is not measurable with respect to a scalar abstention score R , then no policy using only R as an answer-versus-abstain interface is source-complete. Such a score may support selective answering under its assumptions, but it cannot identify the failure source or decide whether a generic correction is safe.*

Proof. A policy using only R can condition only on the information contained in R . If G is not measurable with respect to R , then no function of R can recover the grounding or failure-source state. If A^* is not measurable with respect to R , then no function of R can determine the safe correction action. Therefore, a scalar answer-versus-abstain interface can be selective without being source-complete. $\square$

Implication. The corollary connects scalar abstention to source incompleteness. A scalar score may support selective answering under its assumptions, although it cannot by itself identify the failure source, establish grounding, or decide whether a generic correction is safe.

B Dataset Details

We use four public multimodal benchmarks that stress different sources of prediction risk. The goal is not to build a single pooled hallucination leaderboard. We instead need a dataset suite in which visual fragility, language-prior reliance, and cross-modal alignment instability can be observed under different task conditions. HallusionBench contributes visual illusion and grounded hallucination cases, POPE contributes object-presence hallucination probes, Visual Spatial Reasoning (VSR) contributes explicit relation-binding examples, and VizWiz-VQA contributes real-user visual insufficiency and answerability cases. This mixture supports the source-signature view because the same scalar confidence value can occur in examples with different underlying diagnostic stresses.

B.1 Dataset Roles

Table 6 summarizes the datasets, splits, sample counts, and their roles in the paper. The counts are reported per model before pooling across the four evaluated model families. HallusionBench contains 1,129 datapoints, out of which 178 are text-only. Since HalluPrism requires visual perturbation, we evaluate the 951 image-paired examples. POPE contributes 9,000 object-presence examples across

Dataset	Split / Subset	N Per Model	Primary Diagnostic Stress	Role In HalluPrism
HallusionBench	Image-paired subset	951	Visual illusion, attribute error, and grounded hallucination	Mixed failure-family diagnosis and routing analysis under visually grounded hallucination cases.
POPE	Random, popular, and adversarial splits	9,000	Object-presence hallucination and object priors	Object-hallucination diagnosis, language-prior pressure, and abstention boundary check.
VSR	Zero-shot development examples	340	Spatial and relation binding	Relation-swap alignment probing and routing break-rate analysis.
VizWiz-VQA	Validation split	4,319	Visual insufficiency and answerability	Answerability diagnosis, visual-fragility analysis, and abstention boundary check.

Table 6: Dataset details for the HalluPrism evaluation suite. Counts are per model before pooling across the four evaluated model families.

Dataset	Output Space	Incorrect-Prediction Label	Diagnostic Interpretation
HallusionBench	yes/no	Attribute error or object hallucination	Mixed visually grounded hallucination and illusion failure.
POPE	yes/no	Object hallucination	Object-presence error under random, popular, or adversarial object-prior pressure.
VSR	true/false	Spatial or relation error	Cross-modal binding failure over explicit spatial relations.
VizWiz-VQA	Short answer or <i>unanswerable</i>	Answerability failure	Visual insufficiency or failure to respect answerability constraints.

Table 7: Failure-family mapping used for diagnostic classification. Correct predictions are mapped to *no failure*. Incorrect predictions are mapped using each benchmark’s primary stress factor and available labels.

random, popular, and adversarial splits. VSR contributes 340 zero-shot development examples with explicit spatial relations. VizWiz-VQA contributes 4,319 validation examples and is used for visual insufficiency and answerability analysis.

B.2 HallusionBench

HallusionBench evaluates visually grounded hallucination and illusion cases. We use the image-paired subset because the source-signature protocol requires image perturbations. Text-only examples are excluded since visual degradation, blank-image replacement, and image-grounded alignment probes would not be meaningful without an associated image. The evaluated subset contains yes/no questions, which allows answer normalization into canonical yes and no labels. We map the benchmark categories into attribute-level and object-level hallucination labels for failure-family diagnosis. HallusionBench is useful for the main story because it contains mixed visual and semantic pressures: a model may answer from a plausible prior, from partial visual evidence, or from unstable image-text binding.

B.3 POPE

POPE provides object-presence hallucination probes. We use all 9,000 examples per model across the random, popular, and adversarial splits. The task is binary, and outputs are normalized to yes and no. POPE is primarily used to measure object-hallucination behavior under strong object-prior pressure. It is also useful for the abstention boundary check because object-presence examples often produce high scalar confidence even when language-prior reliance is high. Since POPE does not contain explicit spatial relations, the relation-swap alignment probe is not used. Alignment is measured through the grounding re-check probe.

B.4 Visual Spatial Reasoning

VSR evaluates whether an image satisfies a spatial caption. We use 340 zero-shot development examples. The base prompt asks the model whether the caption is true or false, and outputs are normalized to true and false. VSR is the clearest dataset for relation-sensitive alignment probing because captions contain explicit spatial relations such as *left*, *right*, *above*, *below*, *in front of*, and *behind*. For

Dataset	Per-Model N	Pooled N	Failure-Family Counts Per Model
HallusionBench	951	3,804	attribute = 360, object hallucination = 591
POPE	9,000	36,000	object hallucination = 9,000
VizWiz-VQA	4,319	17,276	answerability = 1,385, none = 2,934
VSR	340	1,360	spatial = 340

Table 8: Dataset composition and failure-family coverage. Counts are reported per model and after pooling across the four evaluated model families.

Dataset	Visual Degradation	Blank Image	Grounding Re-Check	Relation Swap
HallusionBench	Yes	Yes	Yes	No
POPE	Yes	Yes	Yes	No
VSR	Yes	Yes	Yes	Yes
VizWiz-VQA	Yes	Yes	Yes	No

Table 9: Probe availability across datasets. Relation-swapped probing is used only when the input contains an explicit perturbable relation.

VSR, we use both alignment probes: (i) grounding re-checking and (ii) relation-swapped caption verification. The relation-swapped probe is central to testing whether the model preserves the same answer when the expected relation should change. VSR therefore anchors the cross-modal binding part of the source signature.

B.5 VizWiz-VQA

VizWiz-VQA contains visual questions from blind or low-vision users. We use the validation split with 4,319 examples. The dataset is important for two reasons. First, many examples contain weak, blurry, poorly framed, or incomplete visual evidence, which directly stresses visual fragility. Second, some questions are not answerable from the image, making the dataset useful for answerability diagnosis. We instruct the model to return unanswerable when the image is too unclear or the question cannot be answered from visual evidence. We normalize answer variants before exact comparison. Since VizWiz questions are open-ended and do not usually contain controlled spatial relations, the relation-swap probe is not used. Alignment is measured through grounding re-checking.

B.6 Failure-Family Mapping

Each model output receives a correctness label and a failure-family label. Correct predictions are mapped to *no failure*. Incorrect predictions are mapped according to the benchmark stress factor and the available metadata. This mapping is intentionally coarse (see Table 7). We use it to test whether source signatures diagnose broad failure

families, not to claim exhaustive error-taxonomy coverage.

B.7 Probe Availability By Dataset

We use visual degradation, blank-image replacement, and grounding re-checking for all image-paired datasets. We use relation-swapped probing only when the input contains an explicit perturbable relation, which applies to VSR (see Table 9).

B.8 Pooled Evaluation Units

The basic evaluation unit is a model-output instance (x, M) , where x is a dataset example and M is an evaluated model family. We use this pooled unit for source-profile summaries, dominant-source counts, failure-family diagnosis, abstention scalarization, and routing analysis. The per-dataset and pooled counts are reported in Table 8. We also report dataset-identity controls and within-dataset diagnosis because benchmark identity and failure family are partially aligned by design.

C Complete Prompt Catalog

We provide the exact templates used for clean prediction, source-signature probing, probe-level entanglement tests, intervention routing, and auxiliary confidence reporting. These templates define the reproducible prompting protocol used in the experiments.

C.1 Base Dataset Prompts

Base prompts define the clean prediction $\hat{y}$ for each dataset.

**Yes/No Visual Reasoning: POPE and Hallu-
sionBench**

Required Input:

- **Question:** {question}

Prompt Template:

{question}

Answer with only 'Yes' or 'No'.

Spatial Caption Verification: VSR

Required Input:

- **Caption:** {caption}

Prompt Template:

Look at the image carefully. Is the following
statement true or false?

Statement: "{caption}"

Answer with only 'True' or 'False'.

**Open-Ended VQA With Answerability:
VizWiz**

Required Input:

- **Question:** {question}

Prompt Template:

{question}

If the image is too unclear to answer, or the
question cannot be answered from the image,
say 'unanswerable'. Otherwise, give a short
answer (one or two words).

C.2 Diagnostic Source-Signature Probes

These probes produce the perturbed answers used
to compute V , L , and A .

C.2.1 Visual Fragility Probe

The visual probe keeps the base prompt fixed and
applies image degradations.

Visual Fragility Probe

Required Input:

- **Image:** {image}
- **Base Prompt:** {base_prompt}

Image Perturbations:

- Gaussian blur with $\sigma = 15$.

- Center crop covering 50% of each image
dimension, resized back to the original
size.
- Brightness reduction to 30% of the origi-
nal image.
- Gaussian noise with $\mu = 0$ and $\sigma = 25$.

Prompt Template:

{base_prompt}

C.2.2 Language-Prior Reliance Probe

The language-prior probe replaces the original im-
age with a blank canvas while keeping the base
prompt fixed.

Language-Prior Reliance Probe

Required Input:

- **Blank Image:** {blank_image}
- **Base Prompt:** {base_prompt}

Image Replacement:

The original image is replaced with a uniform
blank canvas of the same input size.

Prompt Template:

{base_prompt}

C.2.3 Cross-Modal Alignment Probe

The alignment probe uses grounding re-checking
for all datasets and relation-swapped prompting
only for VSR.

Alignment Probe: Grounding Re-Check

Required Input:

- **Original Prompt:** {original_prompt}

Prompt Template:

Look at the image carefully. First, describe
what you see in the image in one sentence.
Then answer the following question.

{original_prompt}

**Alignment Probe: Relation-Swapped
Prompt for VSR**

Required Input:

- **Caption:** {caption}
- **Relation-Swapped Caption:** {rela-
tion_swapped_caption}

Prompt Template:

Look at the image carefully. Is the following statement true or false?

Statement: "{relation_swapped_caption}"

Answer with only 'True' or 'False'.

Relation-swap construction. For VSR, we construct relation-swapped captions using dataset relations such as *left*, *right*, *above*, *below*, *in front of*, and *behind*.

C.3 Probe-Level Entanglement Stress Tests

These probes test whether targeted perturbations produce diagonal or off-diagonal responses in the measured source signature.

Language-Ruined Probe

Required Input:

- **Decoy Question:** {decoy_question}
- **Answer Constraint:** {constraint}

Example Decoy Question:

What is the capital city of France?

Prompt Template:

{decoy_question}
{constraint}

Alignment-Ruined Probe: Fallback Opposite-Answer Prompt

Use Case:

This fallback is used when deterministic relation inversion or polarity inversion is unavailable.

Required Input:

- **Original Prompt:** {original_prompt}

Prompt Template:

Answer the OPPOSITE of what you would normally answer to the following question.
{original_prompt}

C.4 Intervention Routing Policies

We provide the full prompt and parameter catalog for intervention routing. After computing the normalized dominant source $s_{\text{norm}}^*(x, M)$, we choose one of three matched correction paths: visual enhancement for visual-dominant cases, anti-prior prompting for language-prior-dominant cases,

and grounding or relation checking for alignment-dominant cases. These templates and parameters define the intervention implementation used in the routing results. Appendix D.10 reports only the empirical outcomes.

C.4.1 Generic Intervention Baseline

The generic baseline applies the same correction prompt to every example.

Generic Reasoning Intervention

Required Input:

- **Original Prompt:** {original_prompt}

Prompt Template:

Think step by step about the image and question before answering.
{original_prompt}

C.4.2 Matched Intervention for Visual-Dominant Cases

For visual-dominant cases, we apply image enhancement and keep the original prompt fixed.

Visual-Dominant Intervention: Image Enhancement

Required Input:

- **Image:** {image}
- **Original Prompt:** {original_prompt}

Image Enhancement:

- Upscale the image by $1.5\times$ using LANCZOS interpolation.
- Increase color saturation by $1.1\times$.
- Increase sharpness by $2.5\times$.

Prompt Template:

{original_prompt}

C.4.3 Matched Intervention for Language-Prior-Dominant Cases

For language-prior-dominant cases, we add an anti-prior instruction before the original prompt.

Language-Prior-Dominant Intervention: Anti-Prior Prompt

Required Input:

- **Original Prompt:** {original_prompt}

Prompt Template:

Look at the image very carefully before answering. Do NOT guess based on common sense or what is usually true. Base your answer ONLY on what you can actually see in this specific image.

{original_prompt}

C.4.4 Matched Intervention for Alignment-Dominant Cases

For alignment-dominant cases, we use grounded answering for general inputs and relation checking for VSR.

Alignment-Dominant Grounded Answering

Intervention:

Required Input:

- **Original Prompt:** {original_prompt}

Prompt Template:

Look at the image carefully. First, describe what you see in the image in one sentence. Then answer the following question based on your description.

{original_prompt}

Alignment-Dominant Intervention: VSR Relation Check

Required Input:

- **Caption:** {caption}

Prompt Template:

Follow these steps:

1. First, list the objects you can see in the image.
2. Then, describe the spatial relationship between them.
3. Finally, is this statement true or false: "{caption}"

Answer with only 'True' or 'False' on the last line.

C.5 Answer Normalization

We apply answer normalization before comparing clean, perturbed, and intervened outputs. For binary yes/no datasets, surface variants such as 'yes', 'Yes.', and 'yes, it is' are mapped to the canonical label yes; analogous variants are mapped to no. For VSR, true and false are the canonical labels.

For VizWiz, unanswerable variants are mapped to a single answerability label, and short open-ended answers are lowercased and stripped of punctuation before exact matching. These normalization rules are applied consistently to clean predictions, diagnostic probes, scalarization comparisons, and intervention-routing evaluations.

D Additional Results

We provide additional result that support the source-signature view. We first report probe-design controls for language-prior saturation and alignment availability, and then specify dataset composition and source-profile statistics. We then examine controlled interventions and their off-target effects. After that, we analyze failure-family diagnosis, dataset-identity controls, human validation, abstention, and intervention routing. The additional tables form a single empirical trace from source prevalence to diagnostic utility. The main pattern is consistent across the appendix. Multimodal source signals are entangled under intervention, yet their joint signature remains informative for failure diagnosis. The same signature should not be treated as a direct scalar confidence correction.

D.1 Language-Prior Saturation Controls

The default language-prior score in Eq. 4 measures whether the clean answer survives after removing image evidence. This design is intentionally high-recall: if the same answer persists under a blank image with comparable confidence, the example should be marked as language-prior sensitive. However, the bounded confidence ratio can also saturate when the blank-image answer matches the clean answer, especially for low clean-answer confidence. We therefore run implementation-level controls to check whether the main conclusions depend on this clipped ratio. These controls do not replace the default score used in the main paper. They test whether language-prior saturation alone explains the diagnostic results.

We compare three variants. First, L_{default} is the bounded score from Eq. 4. Second, L_{match} removes the confidence ratio and keeps only blank-image answer survival:

$$L_{\text{match}} = \mathbf{1}[\text{match}(\hat{y}, \hat{y}^\ell)]. \quad (23)$$

Third, L_{stable} keeps the confidence-ratio interpretation but suppresses low-denominator amplification by replacing the clean-confidence denominator

with a fixed floor τ_c :

$$L_{\text{stable}} = \mathbf{1}[\text{match}(\hat{y}, \tilde{y}^\ell)] \cdot \min\left(\frac{c(\tilde{y}^\ell)}{\max(c(\hat{y}), \tau_c) + \epsilon}, 1\right). \quad (24)$$

We set $\tau_c = 0.5$ in all experiments.

We observe from Table 10 that the raw dominance rate of L drops sharply under the control variants, from 99.7% for the default score to about 51.7% for the match-only and denominator-stable variants. However, the diagnostic result remains stable. Replacing L_{default} with L_{match} or L_{stable} changes AUROC by at most 0.006, while Macro-F1 increases from 0.333 to 0.346 and 0.349. Thus, the main diagnostic pattern is not an artifact of the clipped confidence ratio alone. The default L should be read as an answer-survival marker under image removal, not as a causal source label. This is also why the main paper treats raw dominance as a prevalence statistic and uses the full (V, L, A) vector for diagnosis.

D.2 Alignment Availability Controls

The alignment score in Eq. 5 is availability-aware. The grounding re-check component is available for all datasets, while relation-swap probing is available only when the input contains an explicit perturbable relation. In our evaluation suite, this applies to VSR. Therefore, raw A magnitudes should not be interpreted as directly comparable source intensities across all datasets. We instead use A as a behavioral alignment-stress coordinate and rely on dataset/model controls and within-setting normalization when the score is used across heterogeneous benchmarks.

We run an availability control to test whether the diagnostic value of A is driven by the VSR-only relation-swap branch or by the 0.4/0.6 grounding-relation weighting. This control removes VSR entirely from the pooled diagnosis setting. Since VSR is the only relation-eligible dataset, the no-VSR setting contains no relation-swap examples and no example for which the relation weight can affect A . The comparison therefore asks whether the source signature remains diagnostic when only the non-relation benchmark suite is used.

We observe from Table 11 that removing VSR does not reduce the diagnostic value of the source signature. With all datasets, (V, L, A) improves AUROC over scalar confidence from 0.752 to 0.828 and Macro-F1 from 0.260 to 0.333. After excluding VSR, the gain remains nearly the

same in AUROC and becomes larger in Macro-F1: AUROC improves from 0.761 to 0.839, while Macro-F1 improves from 0.329 to 0.422. This indicates that the main diagnosis result is not an artifact of the relation-swap branch or the particular 0.4/0.6 weighting used when relation-swap probing is available. The relation-swap branch should be read as a stronger task-valid alignment stress for relation-bearing inputs, not as the source of the pooled diagnostic effect.

D.3 Full Source Profiles

We use the full source-profile tables to separate raw prevalence from diagnostic value. We observe from Table 12 that L is high across nearly all model-dataset settings, with means from 0.914 to 1.000 and blank-image match rates from 37.5% to 96.5%. In contrast, V and A show stronger task dependence: VizWiz gives the largest visual-fragility values, while VSR gives the largest alignment values for Gemma 4 and Qwen3-VL. We observe from Table 13 that raw $\arg \max(V, L, A)$ selects L for 58,287/58,440 pooled instances. This saturation makes raw dominance a poor proxy for diagnostic value, motivating our use of the full (V, L, A) vector for failure-family diagnosis and normalized dominance for routing.

D.4 Interventional Entanglement Across Models

We next test whether targeted interventions isolate the intended source score. This test is stricter than source profiling. If the three scores behaved as independent source components, visual-ruined inputs would produce the largest absolute change in V , language-ruined inputs would produce the largest absolute change in L , and alignment-ruined inputs would produce the largest absolute change in A . We observe from Table 14 that this magnitude-based diagonal-dominance pattern holds only partially. Across all model-dataset-intervention checks, 18 of 48 pass. Visual-ruined inputs pass in 11 of 16 settings, while language-ruined inputs pass in 2 of 16 and alignment-ruined inputs pass in 5 of 16. This supports a more precise interpretation of (V, L, A) . Visual ruin is often source-selective in magnitude, although the direction is not always positive. Language and alignment interventions remain much less source-selective. Therefore, the vector should be read as a joint behavioral signature rather than as a clean causal decomposition.

Variant	Removal Input	Ratio Treatment	L Dom.	AUROC	Macro-F1
L_{default}	Blank	Clipped ratio	99.7%	0.828	0.333
L_{match}	Blank	None	51.7%	0.823	0.346
L_{stable}	Blank	Floored denominator	51.6%	0.823	0.349

Table 10: Language-prior saturation controls. L Dom. denotes the fraction of examples for which L is the raw dominant coordinate. AUROC and Macro-F1 use the same pooled failure-family diagnosis setting with a stratified 5-fold logistic-regression classifier, replacing only the language-prior coordinate.

Feature Set	Evaluation Scope	Relation-Swap Eligible	AUROC	Macro-F1
c	All datasets	Yes	0.752	0.260
(V, L, A)	All datasets	Yes	0.828	0.333
c	Excluding VSR	No	0.761	0.329
(V, L, A)	Excluding VSR	No	0.839	0.422

Table 11: Alignment availability control. We compare scalar confidence c with the source signature (V, L, A) under the same stratified 5-fold logistic-regression failure-family diagnosis setting. The no-VSR setting removes the only relation-swap-eligible dataset, so any remaining gain cannot be attributed to the VSR-only relation branch or to the grounding-relation weight in Eq. 5.

The model-specific tables show how off-target responses arise across backbones. We observe from Table 15 that Gemma 4 satisfies magnitude-based dominance for visual-ruined inputs in 3 of 4 settings, but the signed response is not always intuitive. On VizWiz, visual ruin reduces V by 0.198 rather than increasing it. On HallusionBench, visual ruin also increases A by 0.126, which shows that visual corruption can change alignment behavior. These rows separate two effects: the intended coordinate may change most in magnitude, while the sign and off-target movement still reveal non-monotone source behavior.

For LLaVA-Mistral, we observe from Table 19 that visual-ruined inputs pass in 3 of 4 settings, while language-ruined inputs pass in none. VizWiz language ruin increases A by 0.255, and VSR visual ruin changes both V and A in the negative direction. These responses indicate that entanglement is not limited to one model family. They also show why magnitude and signed direction should be interpreted separately.

For LLaVA-Vicuna, we observe from Table 17 that only 2 of 12 checks pass overall. HallusionBench alignment ruin passes, while most other checks fail the magnitude-based criterion. POPE visual ruin increases A by 0.239, even though the intended target is visual. VSR language ruin also increases A by 0.197. These rows reinforce the joint-signature interpretation because off-target alignment responses can dominate or substantially accompany the intended response.

For Qwen3-VL, we observe from Table 18 that 7

of 12 checks pass, including all four visual-ruined settings. However, the signed deltas still show non-monotone behavior. On HallusionBench, visual ruin reduces V by 0.085 rather than increasing it, while language ruin reduces L by 0.086 and increases A by 0.175. On POPE, language ruin reduces L by 0.149 and increases A by 0.293. These cases show why independent source attribution is too strong. The vector records how the answer changes under controlled pressure, and a ruined condition can shift the model toward a different answering route.

D.5 Source Signatures By Failure Family

We connect source signatures to failure families before turning to predictive diagnosis. Table 20 reports pooled source means by raw failure-family label. Object hallucination is associated with very high language-prior reliance, with $\bar{L} = 0.996$, and a high scalar score of 0.967. Answerability cases also have near-saturated language-prior scores, with $\bar{L} = 0.997$, although their accuracy is lower at 77.4%. The no-annotated-family row has the highest mean visual and alignment values, with $\bar{V} = 0.289$ and $\bar{A} = 0.587$, because this row is heavily shaped by answerable VizWiz instances. This row should not be read as the correctness-conditioned no-failure target used in the outcome-family controls. These values explain why single-coordinate interpretations are insufficient. Failure families do not map cleanly onto one isolated source. Instead, they occupy different regions of the joint source-signature space.

Model	Dataset	N	Acc	$\bar{V}$	$\bar{L}$	$\bar{A}$	LP Match	Dom- V	Dom- L	Dom- A
Gemma 4	HallusionBench	951	68.2%	0.122	0.958	0.501	56.9%	0	949	2
Gemma 4	POPE	9000	86.9%	0.114	0.999	0.336	53.3%	0	9000	0
Gemma 4	VizWiz	4319	54.0%	0.303	0.996	0.638	37.5%	2	4308	9
Gemma 4	VSR	340	71.8%	0.143	0.999	0.675	64.4%	0	340	0
LLaVA-Mistral	HallusionBench	951	52.6%	0.090	0.979	0.241	58.4%	0	951	0
LLaVA-Mistral	POPE	9000	89.1%	0.104	0.999	0.061	55.8%	0	9000	0
LLaVA-Mistral	VizWiz	4319	54.9%	0.192	0.994	0.479	44.0%	0	4302	17
LLaVA-Mistral	VSR	340	70.3%	0.158	1.000	0.099	50.6%	0	340	0
LLaVA-Vicuna	HallusionBench	951	47.7%	0.071	0.945	0.208	71.2%	0	951	0
LLaVA-Vicuna	POPE	9000	88.6%	0.116	0.992	0.204	56.5%	0	9000	0
LLaVA-Vicuna	VizWiz	4319	53.9%	0.215	0.977	0.604	39.6%	0	4198	121
LLaVA-Vicuna	VSR	340	51.5%	0.037	0.914	0.105	96.5%	0	340	0
Qwen3-VL	HallusionBench	951	74.4%	0.142	0.962	0.318	58.6%	0	951	0
Qwen3-VL	POPE	9000	88.9%	0.094	1.000	0.265	56.9%	0	9000	0
Qwen3-VL	VizWiz	4319	61.3%	0.234	0.999	0.488	39.7%	0	4318	1
Qwen3-VL	VSR	340	87.4%	0.142	0.995	0.613	51.8%	0	339	1

Table 12: Full source-profile with accuracy and raw dominant-source (**Dom**) counts. LP Match denotes exact-answer agreement between the original image and blank-image condition.

Model	N	Dominant V	Dominant L	Dominant A
Gemma 4	14,610	2 (0.01%)	14,597 (99.91%)	11 (0.08%)
LLaVA-Mistral	14,610	0 (0.00%)	14,593 (99.88%)	17 (0.12%)
LLaVA-Vicuna	14,610	0 (0.00%)	14,489 (99.17%)	121 (0.83%)
Qwen3-VL	14,610	0 (0.00%)	14,608 (99.99%)	2 (0.01%)
Pooled	58,440	2 (< 0.01%)	58,287 (99.74%)	151 (0.26%)

Table 13: **Dominant-source Counts.** We compute the raw dominant source using $\arg \max(V, L, A)$. Raw dominance is overwhelmingly language-prior because L is near saturation in most settings; this count should therefore be read as prevalence, not diagnostic value.

[t]

D.6 Full Failure-Family Diagnosis

We report model-wise failure-family AUROC in Table 21. The table compares scalar confidence, the source signature (V, L, A), and the combined representation (V, L, A, c) across classifier families. We report tree-based feature importance in Table 22. These results provide the full supporting evidence for the main diagnosis claim summarized in Section 5.

[t]

D.7 Dataset-Identity And Within-Dataset Controls

The evaluation suite is deliberately heterogeneous: POPE primarily stresses object hallucination, VSR stresses spatial and relation errors, VizWiz stresses answerability, and HallusionBench contributes mixed hallucination cases. This design supports the diagnosis-first claim, although it also means that pooled failure-family prediction can partly reflect benchmark identity. We therefore use dataset-identity, within-dataset, leave-dataset-out, and matched source-stress controls to test whether

source signatures add information beyond coarse dataset and model identifiers.

We observe from Table 23 that dataset and model identifiers already give a strong raw-family baseline. Dataset ID alone reaches AUROC 0.976 and Macro-F1 0.558, while adding scalar confidence raises the result to AUROC 0.986 and Macro-F1 0.705. Adding $V + L + A$ to dataset and model identifiers improves AUROC to 0.991 and Macro-F1 to 0.783. The AUROC gain is small because the controlled baseline is already high, while the Macro-F1 gain indicates better handling of less dominant failure labels.

[t]

We observe from Table 24 that the stricter outcome-family control gives a smaller but positive gain. Here correct predictions are mapped to *no failure*, so the classifier must separate both correctness and error family. Dataset ID and model ID remain weak for Macro-F1, both at 0.145. Adding scalar confidence raises Macro-F1 to 0.308, and the combined representation reaches the best result with AUROC 0.853 and Macro-F1 0.348.

We see in Table 25 that the clearest fixed-dataset

Model	Visual-Ruined	Language-Ruined	Alignment-Ruined	All Checks
Gemma 4	3/4	1/4	1/4	5/12
LLaVA-Mistral	3/4	0/4	1/4	4/12
LLaVA-Vicuna	1/4	0/4	1/4	2/12
Qwen3-VL	4/4	1/4	2/4	7/12
Pooled	11/16	2/16	5/16	18/48

Table 14: **Magnitude-based diagonal-dominance pass counts for the interventional entanglement test.** A check passes when the intended source coordinate has the largest absolute change under the corresponding ruined condition.

Dataset	Intervention	ΔV	ΔL	ΔA	Outcome
HallusionBench	Visual-ruined	-0.075	+0.016	+0.126	FAIL
HallusionBench	Language-ruined	-0.061	-0.025	+0.088	FAIL
HallusionBench	Alignment-ruined	-0.001	-0.000	-0.001	PASS
POPE	Visual-ruined	-0.086	+0.000	+0.068	PASS
POPE	Language-ruined	-0.031	-0.052	+0.247	FAIL
POPE	Alignment-ruined	+0.000	+0.000	+0.000	FAIL
VizWiz	Visual-ruined	-0.198	+0.000	+0.049	PASS
VizWiz	Language-ruined	-0.228	-0.002	-0.009	FAIL
VizWiz	Alignment-ruined	-0.000	+0.000	+0.000	FAIL
VSR	Visual-ruined	-0.143	-0.001	+0.044	PASS
VSR	Language-ruined	-0.023	-0.038	-0.023	PASS
VSR	Alignment-ruined	-0.014	+0.000	+0.005	FAIL

Table 15: **Interventional Entanglement Δ for Gemma 4.** A check passes when the intended source coordinate has the largest absolute change under the corresponding ruined condition. Results are computed using unrounded Δ .

gains appear on HallusionBench and VizWiz. On HallusionBench, scalar confidence reaches AUROC 0.635 and F1 0.558 for attribute-versus-object diagnosis, while (V, L, A) reaches AUROC 0.738 and F1 0.620. The combined representation reaches AUROC 0.774 and F1 0.644. On VizWiz, scalar confidence reaches AUROC 0.715 and F1 0.597, while (V, L, A) reaches AUROC 0.819 and F1 0.745. The combined representation reaches AUROC 0.823 and F1 0.744. POPE and VSR give more conservative fixed-dataset results. On POPE, (V, L, A) is roughly matched with scalar confidence for object-error detection, with AUROC 0.809 versus 0.811, while the combined representation improves to AUROC 0.832 and F1 0.502. On VSR, scalar confidence remains stronger than (V, L, A) in the binary spatial-error setting, with AUROC 0.766 versus 0.732; the combined representation reaches AUROC 0.745. These rows bound the diagnosis claim: source signatures help most when the target contains multiple failure modes or answerability-sensitive distinctions.

We include leave-dataset-out binary transfer as a stress test in Table 26. The results are weak overall, which is expected because the datasets differ in task structure and failure-label support. Holding out POPE gives AUROC 0.201 with model

ID and scalar confidence, 0.375 with model ID and $V + L + A$, and 0.229 with the combined representation. Holding out HallusionBench gives 0.339, 0.492, and 0.371, respectively. We therefore use this table as a transfer stress test, not as the main diagnosis evidence. We find from Table 27 that matched source-stress slices align with the same interpretation. In POPE, language-prior scores remain near saturation across adversarial, popular, and random splits, with $\bar{L}$ between 0.997 and 0.998, while accuracy drops from 90.3% on random to 86.5% on adversarial. In VizWiz, answerable examples have higher visual and alignment scores than unanswerable examples, with $\bar{V} = 0.289$ and $\bar{A} = 0.587$ versus $\bar{V} = 0.124$ and $\bar{A} = 0.479$. VSR relation slices show stable high language-prior scores and moderate alignment scores across relations. These slices provide interpretive support for reading the full (V, L, A) vector rather than a single coordinate.

D.8 Human Audit Results

We report the completed human-audit protocol in Table 28. The audit follows the protocol in Appendix E. We use 200 examples, with 50 examples each from HallusionBench, POPE, VSR, and VizWiz. Three annotators assign one of six la-

Dataset	Intervention	ΔV	ΔL	ΔA	Outcome
HallusionBench	Visual-ruined	-0.026	+0.011	+0.012	PASS
HallusionBench	Language-ruined	-0.051	+0.007	-0.054	FAIL
HallusionBench	Alignment-ruined	+0.001	+0.000	+0.003	PASS
POPE	Visual-ruined	-0.050	+0.001	-0.006	PASS
POPE	Language-ruined	-0.076	-0.014	+0.097	FAIL
POPE	Alignment-ruined	+0.000	+0.000	+0.000	FAIL
VizWiz	Visual-ruined	-0.061	+0.002	-0.175	FAIL
VizWiz	Language-ruined	-0.072	-0.020	+0.255	FAIL
VizWiz	Alignment-ruined	-0.000	+0.000	-0.000	FAIL
VSR	Visual-ruined	-0.153	-0.002	-0.093	PASS
VSR	Language-ruined	-0.146	-0.001	+0.004	FAIL
VSR	Alignment-ruined	-0.023	-0.000	+0.008	FAIL

Table 16: **Interventional Entanglement Δ for LLaVA-Mistral.** A check passes when the intended source coordinate has the largest absolute change under the corresponding ruined condition. Results are computed using unrounded Δ .

Dataset	Intervention	ΔV	ΔL	ΔA	Outcome
HallusionBench	Visual-ruined	-0.020	+0.038	+0.022	FAIL
HallusionBench	Language-ruined	-0.053	+0.037	-0.024	FAIL
HallusionBench	Alignment-ruined	+0.002	-0.001	-0.002	PASS
POPE	Visual-ruined	-0.029	+0.008	+0.239	FAIL
POPE	Language-ruined	-0.101	-0.004	-0.018	FAIL
POPE	Alignment-ruined	+0.000	+0.000	+0.000	FAIL
VizWiz	Visual-ruined	-0.121	+0.016	-0.025	PASS
VizWiz	Language-ruined	-0.061	+0.019	+0.097	FAIL
VizWiz	Alignment-ruined	+0.000	-0.000	-0.000	FAIL
VSR	Visual-ruined	-0.025	+0.074	-0.062	FAIL
VSR	Language-ruined	-0.024	+0.074	+0.197	FAIL
VSR	Alignment-ruined	+0.001	+0.007	-0.001	FAIL

Table 17: **Interventional Entanglement Δ for LLaVA-Vicuna.** A check passes when the intended source coordinate has the largest absolute change under the corresponding ruined condition. Results are computed using unrounded Δ .

bels: visual evidence insufficiency, language-prior reliance, alignment error, mixed, unclear, or no failure. Annotators see the image, question or caption, reference answer, model answer, and correctness flag. They do not see V , L , A , scalar confidence, raw dominant source, normalized dominant source, or routing decision. We manually adjudicate 36 disagreement cases before computing the final audit labels.

We report annotator agreement and adjudicated label counts in Table 29. We read the audit as bounded semantic support for the source-signature view. Full-label Fleiss’ κ is 0.236, and source versus non-source Fleiss’ κ is 0.480. Exact all-annotator agreement is 20.5%, while at least two annotators agree in 82.0% of examples. These values indicate fair full-label agreement and stronger agreement when the task is reduced to source versus non-source attribution. This behavior is expected under the paper’s interpretation, since visual evidence, language priors, and cross-modal binding

can interact in the same example.

After adjudication, 107 examples receive a clear source label: 71 alignment, 20 language-prior, and 16 visual. The remaining cases are 87 unclear and 4 no-failure, and 2 mixed cases after adjudication. Among the 140 originally incorrect examples, the adjudicated labels are 71 alignment, 32 unclear, 19 language-prior, 16 visual, and 2 mixed. This distribution differs sharply from raw source dominance, which is almost always language-prior. It instead matches the main diagnostic pattern in the paper: language-prior is the most prevalent measured signal, while alignment is more useful for separating failure behavior.

We compare the adjudicated human label with raw and normalized dominant sources only on clear-source cases. Mixed, unclear, and no-failure cases are reported separately rather than forced into one of the three source coordinates. We observe from Table 30 that raw dominant source has low agreement with human clear-source labels be-

Dataset	Intervention	ΔV	ΔL	ΔA	Outcome
HallusionBench	Visual-ruined	-0.085	+0.011	-0.009	PASS
HallusionBench	Language-ruined	-0.047	-0.086	+0.175	FAIL
HallusionBench	Alignment-ruined	-0.001	+0.000	+0.002	PASS
POPE	Visual-ruined	-0.049	+0.000	+0.024	PASS
POPE	Language-ruined	+0.003	-0.149	+0.293	FAIL
POPE	Alignment-ruined	+0.000	+0.000	+0.000	FAIL
VizWiz	Visual-ruined	-0.114	-0.000	-0.035	PASS
VizWiz	Language-ruined	-0.114	-0.006	+0.052	FAIL
VizWiz	Alignment-ruined	-0.000	+0.000	+0.000	PASS
VSR	Visual-ruined	-0.126	-0.000	+0.049	PASS
VSR	Language-ruined	+0.081	-0.128	-0.000	PASS
VSR	Alignment-ruined	-0.020	+0.000	-0.003	FAIL

Table 18: **Interventional Entanglement Δ for Qwen3-VL**. A check passes when the intended source coordinate has the largest absolute change under the corresponding ruined condition. Results are computed using unrounded Δ .

Dataset	Intervention	ΔV	ΔL	ΔA	Outcome
HallusionBench	Visual-ruined	-0.026	+0.011	+0.012	PASS
HallusionBench	Language-ruined	-0.051	+0.007	-0.054	FAIL
HallusionBench	Alignment-ruined	+0.001	+0.000	+0.003	PASS
POPE	Visual-ruined	-0.050	+0.001	-0.006	PASS
POPE	Language-ruined	-0.076	-0.014	+0.097	FAIL
POPE	Alignment-ruined	+0.000	+0.000	+0.000	FAIL
VizWiz	Visual-ruined	-0.061	+0.002	-0.175	FAIL
VizWiz	Language-ruined	-0.072	-0.020	+0.255	FAIL
VizWiz	Alignment-ruined	-0.000	+0.000	-0.000	FAIL
VSR	Visual-ruined	-0.153	-0.002	-0.093	PASS
VSR	Language-ruined	-0.146	-0.001	+0.004	FAIL
VSR	Alignment-ruined	-0.023	-0.000	+0.008	FAIL

Table 19: **Interventional Entanglement Δ for LLaVA-Mistral**. A check passes when the intended source coordinate has the largest absolute change under the corresponding ruined condition. Results are computed using unrounded Δ .

cause it collapses to language-prior. Raw dominance reaches clear-source accuracy 0.165, Macro-F1 0.094, and Cohen’s $\kappa = 0.000$. Normalized dominance improves over raw dominance, with clear-source accuracy 0.363, Macro-F1 0.307, and Cohen’s $\kappa = 0.053$, although the agreement remains limited. This supports an important distinction. Raw $\arg \max(V, L, A)$ is useful for measuring prevalence, not for assigning a human-like source label.

We also evaluate a small audit-side diagnostic classifier on the 91 clear-source examples. We report this comparison in Table 31. Scalar confidence reaches weighted AUROC 0.462 for clear-source label prediction, while raw (V, L, A) reaches 0.586. Normalized (z_V, z_L, z_A) reaches 0.511, and raw (V, L, A) with scalar confidence reaches 0.545. The result is exploratory because the audit sample is small, although it is aligned with the main failure-family diagnosis result. The joint source signature carries more source-attribution signal than scalar confidence. At the same time, the audit does not

validate (V, L, A) as independent causal sources. It supports the restrained claim used throughout the paper: source signatures are behavioral diagnostic objects, many examples remain genuinely ambiguous, and argmax collapse loses information that is present in the full vector.

D.9 Abstention And Source-Aware Scalarization

We examine whether source signatures can be collapsed into scalar confidence scores. We observe from Table 32 that scalar confidence remains the strongest correctness-ranking signal under the tested max-penalty, weighted, and geometric scalarizations. The pooled AUROC is 0.783 for scalar confidence and 0.767 for the best source-aware scalarization. The same pattern holds across model families, with the largest gap for Qwen3-VL, where AUROC drops from 0.809 to 0.732. This supports the scalarization boundary: the source signature can help diagnose failure family while still weakening abstention when compressed into a penalty-

Failure Family	N	$\bar{V}$	$\bar{L}$	$\bar{A}$	Scalar	Acc
Answerability	5540	0.124	0.997	0.479	0.880	77.4%
Attribute	1440	0.099	0.946	0.302	0.922	64.3%
None	11736	0.289	0.988	0.587	0.836	45.9%
Object Hallucination	38364	0.107	0.996	0.223	0.967	86.5%
Spatial	1360	0.120	0.977	0.373	0.919	70.2%

Table 20: Pooled source-signature means by failure family.

Model	Classifier	Scalar	V+L+A	V+L+A+Scalar	Δ_{VLA-S}
Gemma 4	Logistic Reg.	0.754±0.009	0.921±0.006	0.932±0.006	+0.167
Gemma 4	XGBoost	0.834±0.013	0.965±0.001	0.974±0.002	+0.131
Gemma 4	LightGBM	0.836±0.012	0.965±0.001	0.973±0.002	+0.130
LLaVA-Mistral	Logistic Reg.	0.847±0.003	0.892±0.003	0.907±0.003	+0.045
LLaVA-Mistral	XGBoost	0.860±0.003	0.973±0.002	0.979±0.002	+0.113
LLaVA-Mistral	LightGBM	0.859±0.004	0.973±0.001	0.980±0.002	+0.115
LLaVA-Vicuna	Logistic Reg.	0.868±0.005	0.876±0.005	0.941±0.004	+0.009
LLaVA-Vicuna	XGBoost	0.873±0.004	0.960±0.001	0.981±0.001	+0.087
LLaVA-Vicuna	LightGBM	0.872±0.004	0.960±0.002	0.981±0.000	+0.087
Qwen3-VL	Logistic Reg.	0.730±0.006	0.756±0.005	0.757±0.006	+0.027
Qwen3-VL	XGBoost	0.806±0.007	0.972±0.002	0.976±0.002	+0.166
Qwen3-VL	LightGBM	0.836±0.005	0.972±0.002	0.976±0.002	+0.137
Pooled	Logistic Reg.	0.752±0.002	0.828±0.003	0.873±0.002	+0.077
Pooled	XGBoost	0.781±0.002	0.954±0.002	0.971±0.001	+0.173
Pooled	LightGBM	0.780±0.002	0.953±0.002	0.968±0.002	+0.172

Table 21: Failure-family prediction AUROC by model and classifier. Δ_{VLA-S} denotes the gain of $V + L + A$ over scalar confidence.

adjusted confidence score.

We observe from Table 33 that this pattern is not only a pooled artifact. Source-aware scalarization is usually worse than scalar confidence across model-dataset slices. The largest drop appears for Qwen3-VL on POPE, from AUROC 0.826 to 0.702. HallusionBench also shows clear drops for Gemma 4 and LLaVA-Mistral, from 0.690 to 0.605 and from 0.609 to 0.552, respectively. The main exception is LLaVA-Vicuna on VSR, where AUROC improves from 0.685 to 0.702. These exceptions are useful boundary cases, although the dominant pattern remains negative for naive scalarization.

We observe from Table 34 that scalar confidence is especially stronger in the low-coverage regime. At 5% coverage, scalar confidence has risk 0.018, compared with 0.023 for weighted scalarization, 0.290 for max-penalty, and 0.225 for geometric scalarization. At 10% coverage, scalar confidence has risk 0.033, while max-penalty and geometric scalarization remain much higher at 0.194 and 0.171. These results locate the proper role of (V, L, A) : source signatures are better suited for failure typing and routing than for naive scalar confidence replacement.

D.10 Intervention Routing

We report the full intervention-routing results in Table 35. The comparison uses three conditions: (i) no intervention, (ii) a generic reasoning intervention applied to every example, and (iii) matched routing based on the normalized dominant source signature. The prompt templates and image-enhancement parameters are given in Appendix C.4. Here we focus only on routing outcomes. We observe from Table 35 that matched routing is conditionally useful rather than a universal accuracy booster. In the pooled result, no intervention reaches 76.6% accuracy, generic intervention reaches 75.7%, and matched routing reaches 75.9%. The aggregate accuracy difference is small. The more relevant pattern is break-rate control: generic intervention has a pooled break rate of 4.1%, while matched routing lowers it to 3.9%. The clearest case appears for Qwen3-VL on VSR. Generic intervention reduces accuracy from 87.4% to 76.2% and breaks 15.5% of originally correct predictions. Matched routing reaches 87.6% accuracy and lowers break rate to 2.7%. Gemma 4 on HallusionBench shows the same direction, with matched routing reducing break rate from 15.1% to 4.9% while recovering the no-intervention accuracy level. On LLaVA-Mistral/VizWiz, both in-

Model	Classifier	V	L	A	Scalar Confidence
Gemma 4	XGBoost	0.105	0.305	0.505	0.085
Gemma 4	LightGBM	0.269	0.153	0.292	0.286
LLaVA-Mistral	XGBoost	0.084	0.102	0.688	0.125
LLaVA-Mistral	LightGBM	0.294	0.099	0.331	0.276
LLaVA-Vicuna	XGBoost	0.107	0.178	0.503	0.212
LLaVA-Vicuna	LightGBM	0.211	0.173	0.317	0.299
Qwen3-VL	XGBoost	0.093	0.274	0.580	0.054
Qwen3-VL	LightGBM	0.257	0.153	0.329	0.260
Pooled	XGBoost	0.147	0.200	0.504	0.150
Pooled	LightGBM	0.222	0.146	0.360	0.271

Table 22: Feature importance for tree-based failure-family diagnosis.

Feature Set	AUROC	Macro-F1
Dataset ID	0.976	0.558
Dataset ID + Model ID	0.975	0.558
Dataset ID + Model ID + Scalar	0.986	0.705
Dataset ID + Model ID + V + L + A	0.991	0.783
Dataset ID + Model ID + V + L + A + Scalar	0.991	0.782

Table 23: Dataset-ID controlled failure-family diagnosis using a stratified held-out split. The target is the benchmark-provided failure-family label used in the pooled diagnosis experiment.

Feature Set	AUROC	Macro-F1
Dataset ID	0.735	0.145
Dataset ID + Model ID	0.743	0.145
Dataset ID + Model ID + Scalar	0.847	0.308
Dataset ID + Model ID + V + L + A	0.838	0.308
Dataset ID + Model ID + V + L + A + Scalar	0.853	0.348

Table 24: Dataset-ID controlled outcome-family diagnosis using a stratified held-out split. Correct predictions are mapped to no-failure and incorrect predictions to their observed failure family.

E Human Audit Protocol

We use a small human audit as a semantic plausibility check for the source-signature interpretation. The audit does not validate hidden causal mechanisms inside a Multimodal Large Language Model (MLLM). It only asks whether a human reader, given the image, question, reference answer, model answer, and correctness flag, can assign a plausible dominant failure source that agrees with the normalized source signature. The audit is therefore aligned with the main paper’s diagnosis-first claim: source signatures are behavioral diagnostic objects, not ground-truth causes.

E.1 Audit Objective

The audit checks whether the source labels used by HalluPrism are interpretable at the example level. In particular, we ask whether model outputs with a visually dominant, language-prior-dominant, or alignment-dominant normalized signature correspond to human-plausible explanations of the observed prediction behavior. We do not use the audit to train the diagnostic classifier, tune source scores, select routing prompts, or choose scalarization weights. All main quantitative results are computed without human audit feedback.

E.2 Sampling Strategy

We sample examples from the same evaluated model-dataset outputs used in the main experi-

terventions reduce accuracy relative to no intervention, although matched routing still lowers break rate from 21.4% to 17.4%. These cases show the intended use of routing: reducing harmful overcorrection when a generic prompt is disruptive.

We observe from Table 36 that matched routing beats generic intervention in 10/16 model-dataset settings and reduces break rate in 10/16 settings. It beats no intervention in 8/16 settings. Thus, HalluPrism should not be read as a general mitigation method. Its decision-facing value is conditional actionability: source signatures help decide when one-size-fits-all correction is risky and when a source-matched correction is safer.

[t]

Dataset	Target	N	Feature Set	AUROC	F1
HallusionBench	Attribute vs object	3804	Scalar	0.634	0.621
			$V + L + A$	0.769	0.718
			$V + L + A + \text{Scalar}$	0.801	0.739
VizWiz	Answerability vs answerable	17276	Scalar	0.707	0.693
			$V + L + A$	0.817	0.782
			$V + L + A + \text{Scalar}$	0.825	0.784
POPE	Object error vs no failure	36000	Scalar	0.823	0.839
			$V + L + A$	0.839	0.853
			$V + L + A + \text{Scalar}$	0.860	0.857
VSR	Spatial error vs no failure	1360	Scalar	0.722	0.688
			$V + L + A$	0.724	0.701
			$V + L + A + \text{Scalar}$	0.735	0.709

Table 25: Within-dataset diagnostic controls. Each row fixes the dataset and includes model identity in all feature sets. Improvements from $V + L + A$ therefore cannot be attributed to dataset identity alone.

Held-Out Dataset	Model ID + Scalar	ModelID + V + L + A	ModelID + V + L + A + Scalar
HallusionBench	0.339	0.492	0.371
POPE	0.201	0.375	0.229
VSR	0.290	0.416	0.274
VizWiz	0.300	0.328	0.331

Table 26: Leave-dataset-out binary failure detection. This is a stress test only, because multiclass failure-family support differs across datasets.

ments. The audit sample contains 200 examples, with 50 examples from each benchmark: HallusionBench, POPE, VizWiz-VQA, and Visual Spatial Reasoning (VSR). Within each dataset, we stratify by normalized dominant source $s_{\text{norm}}^*(x, M)$ whenever possible. This choice is necessary because raw dominance is strongly affected by language-prior saturation, as shown in Appendix D. When a source stratum has fewer available examples, we include all available examples from that stratum and fill the remaining slots with examples sampled from the other normalized-source strata. We also maintain a mixture of correct and incorrect predictions, since the audit checks source plausibility rather than only error explanation.

E.3 Annotator Recruitment And Payment

The human audit was conducted by student annotators recruited locally from undergraduate engineering programs. The annotators were in the 18-22 age group and were studying Computer Science or Electrical Engineering. We selected annotators with sufficient technical background to understand image-question answering tasks, model outputs, correctness flags, and the distinction between visual evidence, language-prior reliance, and alignment-related errors. No annotator was asked to provide personal, demographic, medical, behav-

ioral, or otherwise sensitive information beyond the minimal background needed to report the audit setting. Annotators were not compensated in cash. The compensation arrangement was communicated before participation. The audit involved semantic labeling of benchmark examples and model outputs rather than user-facing data collection or evaluation of the annotators themselves. Annotators were not shown scalar confidence, source scores (V, L, A), raw or normalized dominant sources, routing decisions, or model-derived explanations. This separation reduces score-induced bias and keeps the audit focused on visible example-level evidence.

E.4 Annotator View

Annotators see only information that would be available in a semantic error inspection setting. They are not shown scalar confidence, source scores (V, L, A), raw dominant source, normalized dominant source, routing decision, or any model-derived explanation. This prevents the audit from simply reproducing the model-derived signature.

E.5 Annotation Labels

Each annotator assigns one label from six categories. The first three labels correspond to the source-signature coordinates. Two labels preserve ambiguity rather than forcing a clean source as-

Dataset	Slice	Expected Source	N	Acc	$\bar{V}$	$\bar{L}$	$\bar{A}$	LP Match
POPE	adversarial	<i>L</i>	12000	86.5%	0.119	0.998	0.219	53.8%
POPE	popular	<i>L</i>	12000	88.3%	0.108	0.998	0.209	55.6%
POPE	random	<i>L</i>	12000	90.3%	0.094	0.997	0.222	57.5%
VizWiz	unanswerable	<i>V</i> /answerability	5540	77.4%	0.124	0.997	0.479	77.9%
VizWiz	answerable	<i>V</i> /answerability	11736	45.9%	0.289	0.988	0.587	22.4%
HallusionBench	not visual-required	<i>V</i> / <i>A</i>	2016	56.3%	0.110	0.961	0.318	61.3%
HallusionBench	visual-required	<i>V</i> / <i>A</i>	1788	65.8%	0.102	0.961	0.317	61.2%
VSR	all relations	<i>A</i>	1360	70.2%	0.120	0.977	0.373	65.8%
VSR	relation=at the right side of	<i>A</i>	76	65.8%	0.143	0.982	0.345	59.2%
VSR	relation=behind	<i>A</i>	120	71.7%	0.110	0.980	0.368	66.7%
VSR	relation=facing	<i>A</i>	72	56.9%	0.133	0.984	0.369	70.8%
VSR	relation=in front of	<i>A</i>	104	61.5%	0.130	0.979	0.382	65.4%
VSR	relation=on	<i>A</i>	96	83.3%	0.118	0.967	0.380	58.3%
VSR	relation=on top of	<i>A</i>	72	73.6%	0.131	0.978	0.369	62.5%
VSR	relation=touching	<i>A</i>	80	71.2%	0.116	0.982	0.382	68.8%
VSR	relation=under	<i>A</i>	128	71.9%	0.099	0.972	0.385	75.0%

Table 27: Matched source-stress slices from existing metadata. These slices check whether source signatures align with dataset-specific stress factors under fixed task conditions.

Dataset	<i>N</i>	Raw Agreement	Fleiss’s κ	Model-Human Agreement
HallusionBench	50	84.0%	0.290	45.5%
POPE	50	84.0%	0.201	22.2%
VizWiz	50	62.0%	0.086	37.5%
VSR	50	98.0%	0.221	37.1%
All	200	82.0%	0.236	36.3%

Table 28: Human-audit reporting template. Annotators label the dominant failure source as visual, language-prior, alignment, mixed, unclear, or no-failure. We report agreement (Fleiss’s κ) before adjudication and model-human agreement after adjudication.

Quantity	Value
Examples	200
Annotators	3
Adjudicated disagreement cases	36
Exact all-annotator agreement	20.5%
At least two annotators agree	82.0%
Full-label Fleiss’ κ	0.236
Source vs non-source Fleiss’ κ	0.480
Alignment	71
No failure	4
Unclear	87
Language-prior	20
Visual	16
Mixed	2

Table 29: Human-audit agreement and adjudicated label distribution. The audit uses 200 examples, with 50 examples per dataset.

Human Label	Predicted <i>V</i>	Predicted <i>L</i>	Predicted <i>A</i>
Visual	6	7	3
Language-prior	10	6	4
Alignment	18	26	27

Table 30: Human-audit confusion-matrix template for clear-source examples.

E.6 Annotation Instructions

Annotators are instructed to label the most plausible source of risk in the specific example. They should not infer a model’s hidden reasoning process. They should also avoid using demographic, cultural, or identity-based assumptions not required by the task. For correct predictions, annotators assign a source label only when the prediction appears fragile or source-dependent from the visible evidence. If no source-sensitive risk is visible, they use *Unclear*. This convention keeps the audit aligned with HalluPrism: the source signature is defined for both correct and incorrect predictions, while the human label remains an example-level diagnostic judgment. The instruction shown to annotators is:

signment, while a final label handles cases with no visible source-sensitive risk. The correctness flag is shown to annotators only as context, while the annotation target remains the plausible source of risk or source-sensitive behavior in the example.

Feature Set	Macro-F1	Balanced Acc.	Weighted AUROC
Scalar confidence	0.146	0.229	0.430
Raw V, L, A	0.328	0.330	0.597
Normalized z_V, z_L, z_A	0.269	0.310	0.556
Raw V, L, A + scalar	0.322	0.313	0.587

Table 31: Human audit-side clear-source prediction. The classifier is evaluated on the 107 examples adjudicated as visual, language-prior, or alignment.

Model	Scalar	Penalty	Weighted	Geometric	Best SA
Gemma 4	0.809	0.738	0.773	0.741	0.773
LLaVA-Mistral	0.836	0.718	0.806	0.719	0.806
LLaVA-Vicuna	0.828	0.665	0.778	0.678	0.778
Qwen3-VL	0.809	0.721	0.732	0.721	0.732
Pooled	0.783	0.638	0.767	0.640	0.767

Table 32: Aggregate AUROC for all scalarization variants.

You are given an image, a question or caption, a reference answer, a model answer, and whether the model answer is correct. Your task is to judge the most plausible source of risk or source-sensitive behavior in this example. Use *Visual* if the relevant visual evidence appears weak or insufficient. Use *Language-prior* if the answer seems plausible from the question alone or from common world knowledge rather than the image. Use *Alignment* if the image contains the relevant entities but the answer appears to fail due to relation, attribute, grounding, or image-text binding. Use *Mixed* if more than one source is plausible. Use *Unclear* if the source cannot be determined or if the gold reference answer is factually incorrect. Use *No Failure* if the model answer is completely correct. Do not guess the model’s internal reasoning. Label only the example-level evidence visible to you.

E.7 Agreement and Adjudication

Three annotators independently label each example. We compute raw agreement and Fleiss’s κ over the six-label set: *Visual*, *Language-prior*, *Alignment*, *Mixed*, *Unclear*, and *No Failure*. In addition to the full 6-label agreement, we compute a collapsed Fleiss’s κ that groups the three source labels (Visual, Language-prior, Alignment) against the non-source labels (Mixed, Unclear, No Failure). This isolates whether humans can reliably distinguish source-driven failures from generic ambiguity. Disagreements are adjudicated after the independent pass. During adjudication, annotators may revise a label only by referring to the visible evidence in the image, question, reference answer, model answer, and correctness flag. They do not see source scores during adjudication. After adjudication, we compute model-human agreement between the adjudicated human label and the normalized dominant source $s_{\text{norm}}^*(x, M)$. This com-

parison is restricted to clear-source cases labeled *Visual*, *Language-prior*, or *Alignment*. Mixed and unclear cases are reported separately.

E.8 Reporting Metrics

We report eight audit quantities: (i) *raw agreement*, the fraction of examples on which annotators assign the same label before adjudication; (ii) Fleiss’s κ , the chance-corrected agreement over the six-label set; (iii) *collapsed Fleiss’s κ* , the chance-corrected agreement for source vs. non-source classification; (iv) *clear-source coverage*, the fraction of adjudicated examples assigned to *Visual*, *Language-prior*, or *Alignment*; (v) *model-human agreement*, the agreement between the normalized dominant source and the adjudicated human label on clear-source examples; (vi) *mixed rate*, the fraction of examples adjudicated as *Mixed*; (vii) *unclear rate*, the fraction adjudicated as *Unclear*; and (viii) *no failure rate*, the fraction adjudicated as *No Failure*. Let $\hat{s}(x) = s_{\text{norm}}^*(x, M)$. Let v_h, l_h, a_h, m_h, u_h , and n_h denote the adjudicated labels *Visual*, *Language-prior*, *Alignment*, *Mixed*, *Unclear*, and *No Failure*, respectively. We define the clear-source label set, clear-source examples, label-coordinate map, and agreement indicator as follows:

$$\begin{aligned}
\mathcal{C}_{\text{src}} &= \{v_h, l_h, a_h\}, \quad \mathcal{X}_c = \{x \in \mathcal{X} : h(x) \in \mathcal{C}_{\text{src}}\}, \\
\psi(v_h) &= V, \quad \psi(l_h) = L, \quad \psi(a_h) = A \\
\delta_x &= \mathbf{1}[\hat{s}(x) = \psi(h(x))] .
\end{aligned} \tag{25}$$

Model	Dataset	Scalar AUROC	Best SA AUROC	Δ_{SA-S}	Scalar Sel@60	Best SA Sel@60
Gemma 4	HallusionBench	0.690	0.605 (weighted)	-0.085	77.5%	73.3%
Gemma 4	POPE	0.779	0.741 (penalty)	-0.038	94.7%	94.7%
Gemma 4	VizWiz	0.734	0.718 (weighted)	-0.016	67.4%	67.7%
Gemma 4	VSR	0.712	0.654 (penalty)	-0.058	81.4%	81.4%
LLaVA-Mistral	HallusionBench	0.609	0.552 (penalty)	-0.057	58.2%	54.9%
LLaVA-Mistral	POPE	0.835	0.807 (weighted)	-0.028	97.5%	95.9%
LLaVA-Mistral	VizWiz	0.720	0.666 (weighted)	-0.054	67.3%	64.5%
LLaVA-Mistral	VSR	0.685	0.680 (weighted)	-0.005	79.9%	80.4%
LLaVA-Vicuna	HallusionBench	0.578	0.552 (penalty)	-0.026	50.7%	48.9%
LLaVA-Vicuna	POPE	0.816	0.810 (weighted)	-0.006	97.1%	96.0%
LLaVA-Vicuna	VizWiz	0.722	0.683 (weighted)	-0.039	66.6%	64.2%
LLaVA-Vicuna	VSR	0.685	0.702 (penalty)	+0.017	60.8%	63.2%
Qwen3-VL	HallusionBench	0.675	0.610 (weighted)	-0.065	81.9%	80.0%
Qwen3-VL	POPE	0.826	0.702 (penalty)	-0.124	97.3%	93.7%
Qwen3-VL	VizWiz	0.706	0.682 (weighted)	-0.024	71.8%	71.2%
Qwen3-VL	VSR	0.802	0.751 (penalty)	-0.051	95.6%	95.1%

Table 33: Per-dataset abstention comparison between scalar confidence and the best source-aware scalarization.

Coverage	Scalar Risk	Penalty Risk	Weighted Risk	Geometric Risk	Best SA Risk
5%	0.018	0.290	0.023	0.225	0.023 (Weighted)
10%	0.033	0.194	0.034	0.171	0.034 (Weighted)
20%	0.050	0.134	0.070	0.128	0.070 (Weighted)
40%	0.081	0.094	0.084	0.094	0.084 (Weighted)
60%	0.099	0.157	0.112	0.157	0.112 (Weighted)
80%	0.157	0.224	0.159	0.224	0.159 (Weighted)
100%	0.234	0.234	0.234	0.234	0.234 (Penalty)

Table 34: Pooled risk-coverage values for scalar confidence and source-aware scalarizations.

The reported rates are then:

$$\begin{aligned}
\text{ClearCov} &= \frac{|\mathcal{X}_c|}{|\mathcal{X}|}, \\
\text{MHAgree} &= \frac{1}{|\mathcal{X}_c|} \sum_{x \in \mathcal{X}_c} \delta_x, \\
\text{MixedRate} &= \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \mathbf{1}[h(x) = m_h], \\
\text{UnclearRate} &= \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \mathbf{1}[h(x) = u_h], \\
\text{NoFailureRate} &= \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \mathbf{1}[h(x) = n_h].
\end{aligned} \tag{26}$$

Here $h(x)$ denotes the adjudicated human label. The map ψ links each clear-source human label to the corresponding source-signature coordinate. We report the confusion matrix over clear-source examples and separately report the number of mixed and unclear cases.

E.9 Interpretation

The audit is intentionally conservative. High model-human agreement would support the semantic plausibility of the source-signature labels. Low agreement would not invalidate the main quantitative

results, because the main claims concern behavioral probe responses, failure-family prediction, scalarization boundary checks, and routing breakage. Instead, low agreement would indicate that the normalized dominant source is not always human-interpretable as a single source. This outcome would still be consistent with the paper’s entanglement claim. Therefore, we treat the audit as an interpretability check on the source-label interface, not as evidence for causal source recovery.

F Computational Cost And Resource Use

We report the computational cost of HalluPrism because the method is a diagnosis layer rather than a single-pass confidence score. Each evaluated prediction requires the clean answer, four visual perturbation answers, one blank-image answer, and one or two alignment-probe answers. Thus, the core source-signature computation uses 7 forward passes for datasets without relation-swap probing and 8 forward passes for Visual Spatial Reasoning (VSR), where explicit relation perturbation is available. This cost is the main practical tradeoff of the method: HalluPrism gives source-aware diagnostic information, but it is not intended as a zero-overhead scalar confidence replacement.

Model	Dataset	N	No Interv.	Generic	Matched	Fix _G	Fix _M	Break _G	Break _M
Gemma 4	HallusionBench	951	68.2%	60.9%	68.9%	9.3%	12.6%	15.1%	4.9%
Gemma 4	POPE	9000	86.9%	86.8%	86.3%	11.9%	10.2%	1.9%	2.3%
Gemma 4	VizWiz	4319	54.0%	51.8%	52.2%	5.9%	7.8%	9.2%	9.9%
Gemma 4	VSR	340	71.8%	72.4%	71.5%	22.9%	8.3%	8.2%	3.7%
LLaVA-Mistral	HallusionBench	951	52.6%	50.9%	53.8%	8.2%	12.0%	10.6%	8.4%
LLaVA-Mistral	POPE	9000	89.1%	88.8%	89.2%	6.8%	8.1%	1.2%	0.8%
LLaVA-Mistral	VizWiz	4319	54.9%	49.7%	51.8%	14.5%	14.3%	21.4%	17.4%
LLaVA-Mistral	VSR	340	70.3%	68.8%	70.6%	5.0%	5.9%	4.2%	2.1%
LLaVA-Vicuna	HallusionBench	951	47.7%	47.9%	50.4%	6.4%	13.3%	6.6%	9.0%
LLaVA-Vicuna	POPE	9000	88.6%	89.1%	89.0%	17.7%	16.0%	1.7%	1.6%
LLaVA-Vicuna	VizWiz	4319	53.9%	54.4%	53.3%	11.5%	9.1%	8.9%	8.8%
LLaVA-Vicuna	VSR	340	51.5%	52.1%	52.6%	1.2%	3.0%	0.0%	0.6%
Qwen3-VL	HallusionBench	951	74.4%	71.7%	74.3%	8.6%	16.5%	6.6%	5.8%
Qwen3-VL	POPE	9000	88.9%	88.5%	87.9%	6.7%	6.7%	1.3%	1.9%
Qwen3-VL	VizWiz	4319	61.3%	60.5%	59.2%	5.2%	6.0%	4.5%	7.3%
Qwen3-VL	VSR	340	87.4%	76.2%	87.6%	18.6%	20.9%	15.5%	2.7%
Pooled	All	58440	76.6%	75.7%	75.9%	9.7%	10.0%	4.1%	3.9%

Table 35: Full intervention-routing results. Generic applies one prompt to all samples. Matched chooses an intervention from the dominant source signature.

Model	Matched > Generic	Matched > No Interv.	Lower Break Rate
Gemma 4	2/4	1/4	2/4
LLaVA-Mistral	4/4	3/4	4/4
LLaVA-Vicuna	2/4	3/4	2/4
Qwen3-VL	2/4	1/4	2/4
Pooled	10/16	8/16	10/16

Table 36: Model-level routing win counts.

F.1 Inference Budget

The evaluation contains 58,440 model-output instances across four datasets and four model families. The source-signature computation uses the clean prediction, four visual-perturbation predictions, one blank-image prediction, and one grounding re-check prediction for every image-paired example. Therefore, HallusionBench, POPE, and VizWiz-VQA use 7 forward passes per model-example. Visual Spatial Reasoning (VSR) additionally uses the relation-swap alignment probe, yielding 8 forward passes per model-example.

Let N_d denote the number of examples in dataset d , and let p_d denote the number of forward passes used for source-signature measurement in that dataset. We compute the core source-signature budget as:

$$N_{\text{core}} = \sum_{m \in \mathcal{M}} \sum_{d \in \mathcal{D}} N_d p_d,$$

$$p_d = \begin{cases} 7, & \text{if relation-swap probing is unavailable,} \\ 8, & \text{if relation-swap probing is available.} \end{cases} \quad (27)$$

In our evaluation, relation-swap probing is available only for VSR. Hence, the core source-

signature budget is:

$$N_{\text{core}} = 4(951 \cdot 7 + 9000 \cdot 7 + 4319 \cdot 7 + 340 \cdot 8) = 410,440. \quad (28)$$

Intervention routing adds two further branches for each model-output instance: one generic intervention and one matched intervention. Since the pooled evaluation contains 58,440 model-output instances, routing adds:

$$N_{\text{route}} = 2 \cdot 58,440 = 116,880. \quad (29)$$

The full diagnosis-plus-routing experiment therefore uses:

$$N_{\text{total}} = N_{\text{core}} + N_{\text{route}} = 527,320 \quad (30)$$

forward passes, excluding optional self-reported confidence baselines and implementation-level retries caused by model-serving failures.

F.2 Training Cost

We do not fine-tune any Multimodal Large Language Model (MLLM). All MLLMs are evaluated

Dataset	Sample Size	Primary Stress Factor
HallusionBench	50	Visual illusion, attribute error, and grounded hallucination
POPE	50	Object-presence hallucination and object priors
VizWiz-VQA	50	Visual insufficiency and answerability
VSR	50	Spatial and relation binding
Total	200	Mixed multimodal failure sources

Table 37: Human-audit sampling plan. The audit uses normalized dominant source for stratification because raw dominance is heavily affected by language-prior saturation.

Category	Shown To Annotators	Hidden From Annotators
Input	Image and question or caption	Perturbed images and probe outputs
Output	Model answer	Scalar confidence c
Reference	Gold answer or answerability label	Source scores (V, L, A)
Outcome	Correctness flag	Raw and normalized dominant source
Metadata	Dataset name	Routing decision and intervention result

Table 38: Information shown to annotators and information hidden during the audit.

with deterministic decoding. The only learned models in the paper are lightweight diagnostic classifiers trained on cached scalar confidence and source-signature features. These classifiers include logistic regression, XGBoost, and LightGBM. Their inputs are four-dimensional at most, namely $[V, L, A, c]$, and their cost is negligible relative to MLLM inference. The diagnostic classifiers are used only for analysis. They are not used to update the evaluated MLLMs.

F.3 Decoding And Runtime Controls

We use greedy decoding with temperature 0, disabled sampling, and a maximum response length of 128 tokens. These choices reduce stochastic variation in the perturbation-response measurements. They also make the compute budget easier to interpret, since repeated generations under the same input are not used as uncertainty samples. This differs from sampling-based uncertainty methods, where multiple generations are required for each clean input before any perturbation is applied.

The source-signature cost scales linearly in the number of probes. If a deployment setting needs only a subset of the diagnostic sources, the cost can be reduced by disabling unused probes. For example, a system concerned only with language-prior reliance can run the clean and blank-image branches. A system concerned with spatial binding can retain the relation-swap branch. In the full paper, we keep all probes active because the central empirical question is whether the joint (V, L, A) signature diagnoses failure families and guides correction.

F.4 Hardware And Wall-Clock Reporting

We report the hardware and wall-clock runtime in Table 42. Exact runtime depends on the serving backend, batch size, model implementation, image resolution, and GPU type. The main reproducibility-critical quantity is therefore the number of forward passes, decoding configuration, model checkpoint, and prompt regime. These are fixed by the protocol and reported in Appendix C and Table 41.

F.5 Resource Use And Environmental Reporting

The experiments are inference-only for the evaluated MLLMs and do not require model fine-tuning. This substantially reduces the resource cost compared with training or adapting large vision-language models. However, HalluPrism still increases inference cost because each prediction is evaluated under multiple perturbation branches. We therefore report forward-pass counts explicitly rather than only reporting the number of original examples. If the final execution environment logs GPU-hours or energy use, we will report those values with the hardware description in Table 42. If such logs are unavailable, we report the fixed forward-pass budget and deterministic decoding settings as the reproducible compute description. This is less informative than a full energy audit, but it avoids inventing carbon or power estimates that were not directly measured.

Label	When To Use	Example Pattern
Visual	The image evidence appears weak, degraded, incomplete, too small, occluded, blurry, dark, or otherwise insufficient for the answer.	The model answers an object question, although the relevant object is too blurred or partially outside the frame.
Language-prior	The answer appears plausible from the question alone, common object priors, or default world knowledge, even when the image evidence is weak or absent.	The model answers “yes” to a common object-presence question where the image does not clearly support the object.
Alignment	The relevant entities are visible, while the answer appears to fail because of relation, attribute, grounding, or image-text binding.	The model detects a cone and a car, although it reverses the left-right relation between them.
Mixed	More than one source is plausibly active, and no single source is clearly dominant.	The image is blurry and the question also invites a common prior; either source could plausibly explain the answer.
Unclear	The example does not provide enough information for a reliable judgment, or the provided gold reference answer is factually incorrect.	The image is ambiguous, or the gold answer states there is a TV when the image clearly only contains a monitor.
No Failure	The model prediction accurately matches the correct gold answer, indicating no failure or hallucination occurred.	The model correctly answers the question based on clear visual evidence without error.

Table 39: Human-audit label rubric. The labels are diagnostic descriptions of the observed example, not causal claims about model internals.

Metric	Meaning
Raw agreement	Fraction of examples with identical independent labels.
Fleiss’s κ	Chance-corrected agreement over the six-label set.
Clear-source coverage	Fraction of examples adjudicated as <i>Visual</i> , <i>Language-prior</i> , or <i>Alignment</i> .
Model-human agreement	Agreement between $s_{\text{norm}}^*(x, M)$ and mapped clear-source human labels.
Mixed rate	Fraction adjudicated as <i>Mixed</i> .
Unclear rate	Fraction adjudicated as <i>Unclear</i> .
Collapsed Fleiss’s κ	Chance-corrected agreement for source vs. non-source classification.
No Failure rate	Fraction adjudicated as <i>No Failure</i> .

Table 40: Human-audit reporting fields. We report mixed and unclear cases separately rather than forcing them into one of the three source coordinates.

F.6 Practical Deployment Tradeoff

The compute cost should be interpreted together with the intended role of the method. HalluPrism is not designed to run as a cheap scalar abstention score for every low-risk query. It is more suitable as an audit-time diagnostic, a high-risk inference-time check, or a routing layer for cases where generic correction may damage correct answers. In latency-critical deployments, the full protocol can be reserved for uncertain, high-impact, or audit-selected examples. This preserves the main contribution: source-aware uncertainty should expose the kind of risk present before a system decides whether to answer or correct.

G Artifact, License, and Data Statement

We report the artifacts used in HalluPrism to make dataset use, model access, and release boundaries explicit. All experiments use public benchmark datasets and publicly accessible model checkpoints

or model families. We do not collect new user images, and we do not redistribute benchmark images or model weights. When redistribution is restricted or unclear, we release only derived metadata, evaluation scripts, prompt templates, normalized predictions, source scores, and aggregate result tables, subject to the original artifact licenses.

G.1 Dataset Artifacts

We use four public multimodal benchmarks under their stated access and license terms. We report the dataset-level license and access summary in Table 43. For HallusionBench, we use only the image-paired evaluation subset and do not redistribute raw images. For POPE and VSR, we use the released benchmark prompts, annotations, and examples subject to the corresponding repository terms. For VizWiz-VQA, we treat the raw images with additional care because they may contain user-provided visual contexts. Across all datasets, any released artifact from this paper should contain

Dataset / Branch	Examples Per Model	Models	Passes Per Example	Forward Passes
HallusionBench	951	4	7	26,628
POPE	9,000	4	7	252,000
VizWiz-VQA	4,319	4	7	120,932
VSR	340	4	8	10,880
Core source-signature measurement	14,610	4	–	410,440
Generic and matched routing branches	14,610	4	2	116,880
Full diagnosis-plus-routing experiment	14,610	4	–	527,320

Table 41: Corrected inference budget for HalluPrism. HallusionBench, POPE, and VizWiz-VQA use 7 passes per model-example. VSR uses 8 passes because relation-swap probing is available. Routing adds two additional branches per model-output instance.

Field	Value
GPU type	NVIDIA RTX A6000 (48 GB)
Number of GPUs	3
CPU and memory	AMD EPYC 7543 32-Core Processor, 324GB
Inference framework	HuggingFace Transformers ($\geq 4.40.0$)
Batch size	1 (sequential per-sample inference)
Elapsed calendar time	≈ 15 days
Total sequential GPU compute hours	369.5

Table 42: Hardware and runtime used for the submitted experiments. Runtime depends on serving backend, batching, model implementation, and image preprocessing, so forward-pass counts in Table 41 remain the primary reproducibility quantity.

only derived metadata, prompt templates, normalized outputs, source scores, and aggregate results unless the original dataset license explicitly permits redistribution of raw images.

G.2 Model Artifacts

We evaluate open-weight and publicly accessible Multimodal Large Language Models (MLLMs). We use these models only for inference. We do not fine-tune, redistribute, or modify their weights. All generated outputs are used for research evaluation of source signatures, abstention scalarization, and intervention routing. A summary of the model artifacts is given in Table 44.

G.3 Derived Artifacts

We create derived artifacts from model outputs and perturbation probes. These include clean predictions, normalized answers, scalar confidence values, source scores (V, L, A), dominant-source labels, failure-family labels, abstention scalarization scores, and intervention-routing outcomes. These artifacts are produced from public benchmark examples and model inference. They do not contain new human-subject data. If released, they should be shared as derived metadata keyed by dataset identifiers rather than as redistributed benchmark images, unless the original dataset license explicitly

permits redistribution.

G.4 License Compliance

We use each artifact under its stated research or open-weight access terms. The paper cites the original dataset and model creators in the main bibliography. The released code will include a license file for our own implementation and will document external dependencies separately from dataset and model licenses. Any user of the released artifacts remains responsible for complying with the original licenses and terms for HallusionBench, POPE, VSR, VizWiz-VQA, Gemma 4, Qwen3-VL, LLaVA-Mistral, and LLaVA-Vicuna.

G.5 Data Sensitivity

VizWiz-VQA contains images and questions originating from blind or low-vision users. Although the dataset is public and licensed for research use, the images may include private environments, personal objects, or accessibility-sensitive contexts. We therefore avoid redistributing raw images and treat VizWiz-derived outputs as evaluation metadata. More generally, multimodal datasets can contain visual details that are more sensitive than the corresponding text prompts. Downstream users should inspect dataset licenses, privacy notes, access terms, and intended-use statements before

Artifact	Type	Use In Paper	License / Access Notes
HallusionBench	Benchmark dataset	Visual illusion, attribute error, grounded hallucination, source-profile diagnosis, and routing analysis	The official HallusionBench repository lists the artifact under the BSD-3-Clause license . We use the image-paired evaluation subset and do not redistribute raw images.
POPE	Benchmark dataset and evaluation code	Object-presence hallucination, blank-image pressure, abstention boundary check, and object-hallucination diagnosis	The official POPE repository provides the benchmark code and data. Its license file lists the artifact under the MIT License. We use released prompts and annotations for evaluation.
VSR	Benchmark dataset	Spatial relation verification and relation-swap alignment probing	The official Visual Spatial Reasoning repository lists the project under the Apache-2.0 license . We use zero-shot development examples with explicit spatial relations.
VizWiz-VQA	Benchmark dataset	Visual insufficiency, answerability diagnosis, and visual-fragility analysis	The VizWiz-VQA dataset page states that the work is licensed under a Creative Commons Attribution 4.0 International License . Since the images can contain user-provided visual contexts, we do not redistribute raw images.

Table 43: Dataset artifacts used in HalluPrism. We use public benchmark datasets and follow the corresponding license and access terms.

reusing any raw images or derived artifacts.

I Ethical Considerations and Responsible Use

G.6 Statement Of Use

We use these artifacts only to evaluate the diagnosis-first claim of HalluPrism. The artifacts are not used to train a new deployed MLLM, infer private attributes, identify people, or make decisions about real users. The human audit protocol, if completed, uses benchmark examples and does not collect new sensitive personal information. Any public release of audit labels should exclude annotator-identifying information and should preserve only example identifiers, adjudicated labels, and aggregate agreement statistics.

We study HalluPrism as a behavioral diagnostic layer for Multimodal Large Language Models (MLLMs). The intended use is model analysis, audit support, and safer correction-path selection. It should not be used as a standalone safety certificate, a causal explanation of model internals, or an automatic deployment approval mechanism. This distinction is central to the paper. HalluPrism measures how an output changes under controlled visual, language-prior, and alignment pressure. It does not prove why the model produced the answer, and it does not guarantee that an answer is safe for a new open-world instance.

H Use Of AI Writing Assistance

AI writing assistance was used during manuscript preparation for language editing and polishing of LATEX tables. The authors remained responsible for all scientific content, including the problem formulation, experimental design, implementation, data processing, result interpretation, citations, mathematical statements, and final manuscript text. The authors manually verified the reported claims, numerical results, tables, citations, proofs, and responsible-use statements before submission. AI assistance was not used as a substitute for author judgment, was not treated as authorship, and did not determine the paper’s scientific conclusions.

I.1 Intended Use

HalluPrism is intended for model auditing, benchmark-level error analysis, and guarded correction-path selection. Appropriate uses include: (i) comparing model behavior across multimodal hallucination benchmarks, (ii) identifying broad failure-family patterns that are not visible from scalar confidence alone, and (iii) selecting a correction path only when validation shows that such routing reduces harmful breakage. Any deployment use should specify the task, user population, validation distribution, acceptable risk level, fallback policy, and monitoring plan before source signatures are used to guide correction.

Artifact	Type	Use In Paper	License / Access Notes
Gemma 4	Open-weight MLLM family	One evaluated model family for source-signature measurement and routing analysis	Google AI documentation describes Gemma 4 as open-weight with responsible commercial use. Use remains subject to the applicable Gemma terms , Gemma Prohibited Use Policy , and the Gemma 4 model documentation .
Qwen3-VL	Open-weight MLLM	One evaluated model family for source-signature measurement, abstention scalarization, and routing analysis	The Qwen3-VL-8B-Instruct model card is used as the checkpoint reference. The model is listed with an Apache-2.0 license ; the Azure/Hugging Face catalog also lists Qwen3-VL-8B-Instruct as Apache-2.0.
LLaVA-Mistral	Open-weight MLLM	One evaluated model family for source-signature measurement and diagnostic controls	The llava-v1.6-mistral-7b-hf model card is used as the checkpoint reference. The model card metadata lists Apache-2.0 as the license.
LLaVA-Vicuna	Open-weight MLLM	One evaluated model family for source-signature measurement and diagnostic controls	The llava-v1.6-vicuna-7b model card is used as the checkpoint reference. The model card states that Llama 2 is licensed under the Llama 2 Community License .

Table 44: Model artifacts used in HalluPrism. We use the listed models for deterministic inference only and do not redistribute model weights.

Derived Artifact	Content	Release Boundary
Prediction files	Clean model answer, normalized answer, scalar confidence, correctness flag, and failure-family label	Can be released with dataset identifiers, subject to dataset terms.
Source-signature files	V , L , A , raw dominant source, normalized dominant source, and probe metadata	Can be released as derived metadata.
Perturbation metadata	Names and parameters for blur, crop, brightness, noise, blank-image replacement, grounding, and relation-swap probes	Can be released as code and metadata.
Routing files	Generic intervention output, matched intervention output, fix indicator, break indicator, and post-intervention accuracy	Can be released as derived metadata if original dataset terms permit.
Prompt catalog	Base prompts, diagnostic probes, entanglement probes, generic intervention, and matched intervention templates	Can be released in full.

Table 45: Derived artifacts produced by the HalluPrism pipeline. These artifacts support reproducibility without requiring redistribution of raw benchmark images or model weights.

I.2 Non-Intended Use

HalluPrism should not be used as a standalone safety certificate, a deployment approval mechanism, or a replacement for human judgment in high-stakes settings. It should not certify answers in medical, legal, financial, educational, accessibility-critical, or public-sector contexts without domain-specific validation, expert review, and post-deployment monitoring. The source scores should also not be interpreted as ground-truth causes of model behavior. They are bounded diagnostic signals produced under a fixed evaluation protocol. Treating them as causal explanations or complete reliability guarantees would overstate what the method establishes.

I.3 Risks From Diagnostic Misuse

The main misuse risk is over-trust. A system that treats the source signature as a complete reliability certificate recreates the same failure mode that motivates the paper’s caution about scalar abstention. The signature is bounded by the evaluated datasets, models, prompts, perturbations, and routing choices. It can miss risks outside these conditions, including domain expertise gaps, cultural context, ambiguous user intent, missing background information, and harms that are not expressed through the measured source coordinates. A second risk is correction harm. A correction prompt can improve one model-dataset condition and damage another. Therefore, routing should be treated as conditional actionability rather than

Use Case	Appropriate Use	Boundary Or Risk
Model auditing	Compare scalar confidence with source signatures and failure-family labels.	Does not recover hidden causal mechanisms.
Selective answering	Use scalar scores only under explicit validation-to-deployment assumptions.	A threshold is a deployment guardrail, not an instance-grounded guarantee.
Correction routing	Use (V, L, A) to choose visual enhancement, anti-prior prompting, or relation checking.	Routing is conditional and may still break correct answers.
High-stakes deployment	Use only with domain-specific validation, monitoring, and human oversight.	Not sufficient for medical, legal, financial, or accessibility-critical certification.
Human audit	Check whether normalized dominant sources are semantically plausible.	Human labels do not prove causal source separability.

Table 46: Responsible-use boundaries for HalluPrism. The method is intended for diagnosis and matched correction analysis, not for standalone safety certification.

universal mitigation. Systems using HalluPrism should log the original answer, routed action, corrected answer, fix indicator, break indicator, and any human-review outcome, since accuracy alone can hide harmful overcorrection.

I.4 Risks For Affected Users

Several datasets used in this paper involve user-facing visual question answering. VizWiz-VQA is especially relevant because it contains images and questions from blind or low-vision users. Errors in such settings may create practical harm because users may not be able to independently verify a visual answer. A model that confidently guesses from a language prior may give a fluent answer that is not supported by the image. A generic correction prompt may also change an originally correct answer. We therefore view HalluPrism as an audit aid for identifying risk structure, not as a substitute for accessibility-aware system design, user consent, domain-specific evaluation, or human fallback mechanisms.

I.5 Data And Privacy Considerations

We use public benchmark datasets and do not collect new user images for the main experiments. The human audit protocol, if completed, uses examples sampled from the evaluated benchmark outputs and does not require collecting personal information from annotators beyond standard annotation logistics. Annotators do not see source scores or routing decisions, which limits score-induced bias during the audit. Public image datasets can still contain sensitive visual content, private environments, or user-specific contexts. Any released artifact should therefore include dataset source information, license terms, access constraints, and a clear description of which fields are original, de-

rived, or model-generated.

I.6 Responsible Deployment Guidance

Before using HalluPrism in deployment, a system owner should document: (i) the target task and user population; (ii) the validation distribution and expected deployment shifts; (iii) the acceptable risk level and fallback policy; (iv) the conditions under which source signatures trigger correction or human review; (v) the logging fields for the original answer, source signature, routed action, corrected answer, fix indicator, and break indicator; and (vi) the monitoring plan for post-deployment drift. In high-impact or latency-critical settings, the full protocol should be reserved for audit-selected, uncertain, or high-risk examples. Source signatures should guide review or guarded correction rather than automatically certifying an answer as safe.